A Study on the Composability Gap in Diffusion Models for Clinical Out-of-Distribution Generation

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Abstract

Compositional generative modeling aims to create complex unseen combinations by composing multiple pretrained models. Instead of relying on a single large-scale system, individual expert models can be combined at inference time to enable out-of-distribution generation. A single model must learn these combinations directly from training data, which becomes difficult when many of them are rare or missing. In contrast, compositional approaches factorize this problem into reusable components, enabling data-efficient generalization to novel combinations without retraining. This capability is critical in low-resource settings where specific data combinations are rare. In clinical imaging, for example, hospitals often have many scans for single diseases but very few examples of rare co-morbidities. A compositional system might bridge this gap by synthesizing plausible rare cases from these separate components to support data augmentation.

However, current compositional methods often fail. These systems struggle with tasks like attribute binding, negation, and combining closely related concepts. A primary reason for this failure is the mathematical operator used to combine conditions, which we term as "logical composition". This mathematical approach does not always match the semantically meaningful output produced by a single prompt for a complex scene. This creates a disconnect between the mathematical combination and the intended scene. The result limits image quality and makes synthetic data unreliable for sensitive clinical use.

The proposed study investigates this disconnect and aims to develop a method to reduce it. The research follows three steps. First, it characterizes the failure of logical composition through a structured categorization of concept pairs and trajectory analysis. Second, it tests whether learning more factorized representations, using a partitioned latent space, can align the mathematical combinations with the target scene. Finally, it demonstrates the utility of this framework in medical imaging. The proposed study evaluates whether rare, co-morbid chest X-rays can be synthesized from individual common and separate disease models. This work aims to make out-of-distribution synthesis more reliable and useful for practical downstream tasks.

Declaration

I, Molefe Reabetsoe-phenyo Motlhabane Molefe, hereby declare the contents of this research proposal to be my own work. This proposal is submitted for the degree of PhD in Computer Science at the University of the Witwatersrand. This work has not been submitted to any other university, or for any other degree.

Signature **Date**

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Chapter 1

Introduction

Diffusion models [Song and Kingma 2021; Ho *et al.* 2020] have become a leading framework for high-fidelity generative modeling. Recent advances [Rombach *et al.* 2022; Betker *et al.*] in text-to-image diffusion models have significantly improved the ability to generate highly realistic images. These capabilities have rapidly extended beyond vision to domains such as speech [Chen *et al.* 2020], medical image reconstruction [Mei *et al.* 2023; Cao *et al.* 2024], and natural language processing [Austin *et al.* 2021; Li *et al.* 2022; Hoogeboom *et al.* 2021]. Together, these developments position diffusion models as a general and increasingly universal framework for generating and manipulating complex data across modalities.

Their strength comes from their iterative sampling process. Instead of producing an output in a single step, a diffusion model starts from noise and refines it through a sequence of denoising updates. This approach makes generation more stable and controllable than earlier architectures like GANs [Goodfellow *et al.* 2014]. This stability helps explain why diffusion models achieve such high sample quality across many diverse domains.

This progress is especially important in medical imaging [Mei *et al.* 2023; Cao *et al.* 2024]. Diffusion models are now used for image denoising [Ho *et al.* 2020], super-resolution [Saharia *et al.* 2023], and synthetic data generation [Rombach *et al.* 2022]. In low-resource clinical settings, these capabilities are particularly valuable because high-quality data are often scarce or expensive to obtain [Gröger *et al.* 2025; Nyamawe and Shao 2026]. For example, hospitals may have many images of common single conditions but very few examples of rare co-morbid pathologies. A generative model that can produce anatomically plausible rare cases could therefore support critical downstream tasks through data augmentation [Holste *et al.* 2024].

One useful way to understand these models is through their connection to energy-based modeling [LeCun *et al.* 2006]. In this view, the score function guides a noisy sample toward regions of high probability. This perspective also suggests a route to compositional generation. If separate models represent different conditions, their scores can be combined at inference time to form a joint generation process. Prior work [Du *et al.* 2023; Du and Kaelbling 2024; Skreta *et al.* 2024; Du *et al.* 2020] has shown that this makes it possible to compose concepts without retraining a new model for every combination.

However, this compositional strategy does not reliably produce the intended result [Huang *et al.* 2025; Bradley *et al.* 2025; Du and Kaelbling 2024]. The "logical composition" defined by the mathematical operators often diverge from the "semantic composition" expressable by a single prompt. In practice, this leads to failures such as incorrect attribute binding and interference between these concepts [Huang *et al.* 2025]. In clinical imaging, these failures are especially serious because they can blur distinct pathologies or produce anatomically implausible scans. This gap between logical and semantic composition is the central problem addressed in the proposed study.

Bradley *et al.* [2025] argue that composition cannot be defined purely in terms of the input distributions, and instead requires specifying which aspects of each distribution should be preserved in the composed result. Their "projective composition" perspective suggests that success depends on whether the relevant factors can be separated in a suitable representation, not only on whether scores are added in the right way. This suggests that the key issue is not only how composition is performed, but also how the underlying

representations in the individual expert models are organised. If the factors remain entangled, the model does not expose the factorisation needed for successful composition.

This reveals a broader unresolved question in compositional generative modeling. Although there is substantial empirical work on controllable diffusion [Wu *et al.* 2024; Xing *et al.* 2024], the link between compositional success and representational factorisation remains underexplored. Recent analyses [Bradley *et al.* 2025; Du and Kaelbling 2024; Huang *et al.* 2025] suggest that compositional failures often arise because the learned representation does not keep the relevant factors sufficiently separate, so composition produces conflicts or incorrect associations instead of the intended semantic combination. Prior work on disentangled representation learning [Locatello *et al.* 2019; Träuble *et al.* 2021b] shows that recovering disentangled factors in unsupervised settings is fundamentally ill-posed without suitable inductive biases. These observations motivate the hypothesis that failures in compositional diffusion may arise from limitations of the learned representation rather than composition alone.

Accordingly, the proposed study pursues three linked objectives and makes three corresponding contributions:

1. **Characterising the composability gap.** The study first develops a dynamics-based measurement framework for analysing how logical score-based composition diverges from semantic composition across different composition regimes. This yields a systematic account of when the gap emerges and identifies the scenarios in which composition produces consistent failures.
2. **Learning representations for composition.** The study then introduces a novel weakly supervised partitioned VAE architecture designed to disentangle the latent space into factors that remain composable under downstream diffusion-based generation. This provides a controlled test of whether improved representational factorisation can reduce the composability gap.
3. **Demonstrating clinical utility for Out-Of-Distribution generation.** The study finally applies this architecture to the synthesis of rare co-morbid chest X-rays from more abundant single-disease data, establishing a clinically meaningful low-resource use case for compositional diffusion. To our knowledge, this is the first study to investigate composable diffusion models for out-of-distribution generation in medical imaging.

These objectives are operationalised through the three research phases summarised in Table 1.1.

Table 1.1: Research phases, objectives, and core assumptions.

Phase	Objective	Hypotheses
Phase 1	Characterise the composability gap by comparing logical and semantic composition across controlled concept-pair regimes.	1. Bayes-style score composition succeeds only when the model already contains a usable approximation to the desired joint configuration. 2. The severity of the composability gap varies systematically with the structure of the underlying concept pair.
Phase 2	Test whether more factorised latent representations reduce the composability gap in a controlled synthetic setting.	1. Known generative factors in dSprites [Matthey *et al.* 2017] and correlated dSprites provide a valid controlled test bed for held-out composition. 2. If representation quality limits composition, more factorised latent spaces should improve held-out composition under a fixed downstream evaluator.
Phase 3	Test whether partitioned clinical representations support plausible low-resource co-morbid chest X-ray generation.	1. Shared anatomical structure and disease-specific effects can be represented as separable components in the latent space. 2. Diseases can be composed from sufficiently factorized latent components.

The remainder of the proposal is organised as follows. Chapter 2 reviews the technical background on VAEs, disentangled representation learning, diffusion models, and compositional diffusion. Chapter 3 situates the proposal within prior work on compositional generation and medical image synthesis. Chapter 4 states the research goals and hypotheses. Chapter 5 presents the three-phase research method used to test them. Chapter 6 outlines the project timeline, success criteria, and expected challenges. Chapter 7 summarises the proposal and its expected contribution.

Chapter 2

Background

This chapter provides the background for the three phases of the study. Section 2.1 introduces variational autoencoders and disentangled representation learning, which motivates Phase 2. Section 2.2 introduces diffusion models, their score-based formulation, Stable Diffusion, and supporting vision-language models used for prompt inversion and text-image alignment. Together with compositional diffusion, these ideas underpin all three phases.

2.1 Variational Autoencoders

A standard autoencoder can learn a compact representation for reconstructing an input, but this representation is not necessarily suitable for generation. A variational autoencoder (VAE) [Kingma and Welling 2013] addresses this limitation by making the latent representation probabilistic and regularising it toward a simple reference distribution. The result is a model that can both reconstruct observed examples and generate new ones from sampled latent codes.

During training, the encoder takes an input image x and produces an approximate posterior distribution $q_\phi(z \mid x)$ over latent variables $z \in \mathbb{R}^d$, where ϕ denotes the encoder's learnable parameters. In the standard Gaussian case, this distribution is parameterised by a mean $\mu_\phi(x)$ and a diagonal covariance $\mathrm{diag}(\sigma^2_\phi(x))$, where $\mu_\phi(\cdot)$ and $\sigma^2_\phi(\cdot)$ are functions defined by the encoder. A latent sample z drawn from this distribution is then passed to the decoder to reconstruct the input. At inference time, we sample $z \sim p(z) = \mathcal{N}(0, I)$ from the prior and decode it directly.

More formally, a VAE defines a latent-variable generative model $p_\theta(x) = \int p_\theta(x \mid z)\, p(z)\, \mathrm{d}z$, where θ denotes the learnable parameters of a model intended to approximate the data distribution, $p(z) = \mathcal{N}(0, I)$ is a standard Gaussian prior, and $p_\theta(x \mid z)$ is a learned decoder. Because the marginal likelihood is intractable, training maximises the evidence lower bound (ELBO) such that the objective function becomes:

$$\mathcal{L}_{\mathrm{VAE}}(\theta, \phi; x) = \underbrace{\mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x \mid z)]}_{\text{reconstruction}} - \underbrace{D_{\mathrm{KL}}(q_\phi(z \mid x) \,\|\, p(z))}_{\text{regularisation}}, \tag{2.1}$$

where $q_\phi(z \mid x) = \mathcal{N}(\mu_\phi(x), \mathrm{diag}(\sigma^2_\phi(x)))$ is the encoder's approximate posterior. The first term rewards reconstructions that explain the observed data and the second keeps the posterior close to the prior.

2.1.1 Disentangled Representations

A representation is *disentangled* if each latent dimension encodes at most one ground-truth factor of variation, and changing one dimension changes the corresponding attribute without affecting others [Wang *et al.* 2024; Träuble *et al.* 2021a].

A representation is *disentangled* if each latent dimension encodes at most one factor of variation in the data, where a factor of variation refers to an attribute whose independent changes produce consistent and interpretable changes in the observed data. In such representations, varying one dimension affects only the corresponding attribute while leaving others unchanged [Wang *et al.* 2024; Träuble *et al.* 2021a]. Disentanglement is desirable because it induces a separable latent structure, where different dimensions

capture distinct factors of variation rather than entangling them. This makes the representation easier to interpret and enables controlled changes in generated data, since one attribute can be varied without unintentionally affecting others.

One common way to encourage this structure is to modify the VAE objective. By scaling the KL term by $\beta > 1$ in Equation 2.1, β-VAE [Higgins *et al.* 2016] restricts how much information can pass through the model. This pressure encourages the encoder to organise data into a set of distinct, unrelated traits. FactorVAE [Kim and Mnih 2018] pursues the same goal through a different modification of the objective, replacing the β multiplier with a Total Correlation penalty ($\gamma\,\mathrm{TC}(q(z))$). This term, $\mathrm{TC}(q(z)) = D_{\mathrm{KL}}(q(z) \,\|\, \prod_j q(z_j))$, specifically measures how much the different features overlap or depend on one another, pushing them to be more independent. Both methods try to force independence within a single shared code $z \in \mathbb{R}^d$. This works when the data's features are independent, but if the training data contains hidden patterns or correlations, the model will use them as a shortcut. In those cases, simply adding more regularization is not enough to prevent the model from following correlations in the data [Locatello *et al.* 2019]. Locatello *et al.* [2019] show that unsupervised disentanglement is unreliable without inductive bias, that is, assumptions in the model or training procedure that favour certain latent structures over others. Without such constraints, different random seeds and hyperparameter settings can lead to inconsistent results, and models that appear well disentangled cannot be selected or reproduced reliably without some form of supervision. Moreover, for the datasets they study, higher disentanglement does not consistently improve downstream sample efficiency.

2.1.2 Weakly Supervised Disentanglement

To address this limitation, inductive bias is introduced in two main ways: weak supervision through factor labels [Locatello *et al.* 2020; Träuble *et al.* 2021a] and structural partitioning of the latent space [Louiset *et al.* 2024]. Both approaches aim to make factors easier to separate under correlated training data, where independence-based objectives alone are often insufficient. These approaches are particularly relevant in settings where compositional generation depends on the ability to isolate and manipulate individual factors within the latent space. In the proposed study, we hypothesize that failures in compositional diffusion arise from entangled representations rather than the composition rule itself, motivating the use of inductive biases that explicitly encourage factor separation.

Weak supervision through training-time predictor heads. The first approach uses training-only predictor heads attached to the encoder representation, as in weakly supervised disentanglement [Locatello *et al.* 2020] and and subsequent work on attribute disentanglement [Scaramuzzino *et al.* 2023]. In this setting, the relevant bias comes from auxiliary supervision, that is, an additional label-based training signal used to shape the representation without being required at inference time. This supervision encourages the encoder to keep selected factors explicitly predictable from the latent code, rather than allowing them to mix arbitrarily with other sources of variation. For each known factor k, a small MLP g_k is applied to the deterministic encoder mean $\mu_\phi(x)$ and trained against the ground-truth label y_k using either a cross-entropy or regression loss, depending on the factor type. These heads are used only during training and are discarded at inference, so the supervision helps organise the representation while leaving the generative model unchanged at inference time. Applying the auxiliary loss to $\mu_\phi(x)$ rather than to a stochastic sample z provides a stable, low-variance training signal and matches the deterministic representation typically used in disentanglement evaluation [Eastwood and Williams 2018; Locatello *et al.* 2019]. Under correlated training data, this form of weak supervision has been shown to recover part of the factor structure that independence-based objectives alone fail to separate [Träuble *et al.* 2021b].

Correct-head supervision and adversarial exclusion. Given a latent representation z, it is common to structure z into subsets intended to represent distinct factors of variation. Auxiliary supervision is then used to encourage each factor to be encoded in its designated subset. However, this does not ensure that the same factor is absent from other subsets [Locatello *et al.* 2020; Scaramuzzino *et al.* 2023]. Without additional constraints, latent representations are not identifiable and may encode the same factor redundantly across multiple components [Locatello *et al.* 2020].

Predictors provide a mechanism to associate subsets of the latent representation with specific factors. A predictor is a neural network that takes a subset of z as input and is trained to predict a corresponding factor label using a classification loss.

For example, a predictor attached to z_{color} encourages color information to be recoverable from that subset, but the same information may remain present in other subsets such as z_{shape}. Applying predictors to these subsets reveals whether factor information is present outside its designated subset. However, using a standard classification loss in this setting encourages the encoder to preserve factor information in those subsets, since accurate prediction requires the factor to remain present [Scaramuzzino *et al.* 2023].

To address this, Ganin *et al.* [2016] introduce the gradient reversal layer (GRL). The GRL, depicted in Figure 2.1, leaves the forward pass unchanged, allowing the predictor to be trained to detect the presence of a factor, but reverses the gradient during backpropagation. As a result, the predictor becomes more accurate at identifying factor information, while the encoder is trained to remove that information from the corresponding representation. This induces an adversarial interaction between the predictor and the encoder.

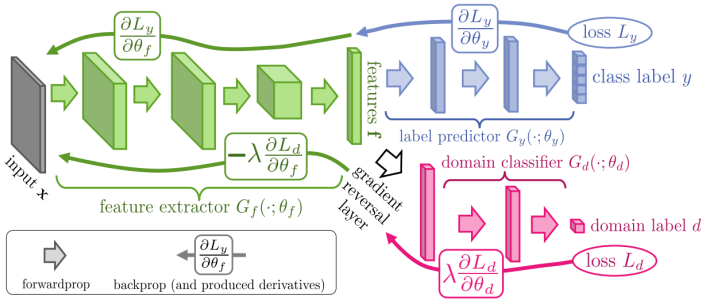


Figure 2.1: Domain-Adversarial Neural Networks (DANN) illustrate the gradient reversal mechanism introduced by Ganin *et al.* [2016]. In DANN, a label predictor is trained for the main task, while an auxiliary domain classifier is attached through a gradient reversal layer so that the learned features become uninformative about the domain. The same mechanism can be reused for factor-specific exclusion: a predictor is attached to a subset of the latent representation where a factor is not intended to be encoded, and the reversed gradient pushes the encoder to remove that factor from that subset.

In this formulation, predictors attached to the intended subset are referred to as *correct-heads*, as they encourage recoverability of the factor from its designated location. Predictors attached to other subsets are referred to as *wrong-heads*. When combined with a gradient reversal layer, wrong-head predictors do not encourage accurate classification, but instead drive the encoder to remove the corresponding factor from those subsets. Together, correct-head supervision and GRL-based wrong-head suppression encourage factor information to be localized to its designated subset while being suppressed in others [Scaramuzzino *et al.* 2023].

Structural latent partitioning across datasets, groups, and modalities. Structural latent partitioning addresses disentanglement at the level of model design by introducing explicit factorisations of the latent space [Bouchacourt *et al.* 2018]. Rather than relying solely on regularisation to separate factors within a shared latent code, these approaches define distinct latent variables with different roles, such as shared and factor-specific components [Chen and Batmanghelich 2019; Suzuki *et al.* 2016]. The shared variables capture invariant structure (e.g. object identity or cross-modal semantics), while the factor-specific variables capture residual variation (e.g. viewpoint, modality-specific features, or style). In multi-view settings, this can be instantiated by introducing a view-common latent variable alongside view-specific latent variables [Xu *et al.* 2021]. The exact split depends on the structure available in the data, for example whether the setting involves grouped observations [Bouchacourt *et al.* 2018] or multiple aligned modalities [Xu *et al.* 2021].

Contrastive VAE is motivated by settings in which the variation of interest is enriched in a target dataset relative to a background dataset [Abid and Zou 2019]. It partitions the latent space into a common subspace, which captures variation present in both datasets, and a salient subspace, which captures variation specific to the target set. Background samples are reconstructed without using the salient code, which pushes target-specific structure into the salient partition.

Multimodal VAEs are motivated by settings in which the same underlying example is observed through more than one modality, such as paired image and text, or image and segmentation mask [Hsu and Glass 2018; Shi *et al.* 2019]. Their latent spaces are therefore divided into a shared component, which captures

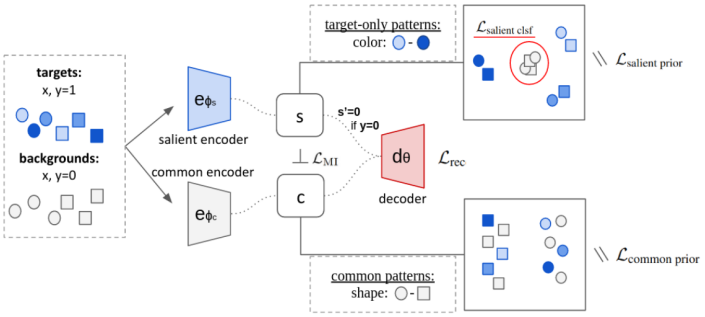


Figure 2.2: SepVAE training pipeline. Background samples ($y = 0$) are forced to use only the common latent space by setting the salient representation to $s' = 0$, while target samples ($y = 1$) can additionally use the salient latent to encode target-specific variation.

information present in all modalities, and private components, which capture variation specific to each modality. This partition is useful for tasks such as joint generation and predicting one modality from another. In practice, however, recovering this separation reliably remains difficult when modality-specific variation dominates [Märtens and Yau 2024].

Louiset *et al.* [2024] introduce SepVAE, depicted in Figure 2.2, a model that implements structural latent partitioning in a contrastive setting by decomposing the latent space into a common component c, capturing variation shared across background and target datasets, and a salient component s, capturing factors present only in a designated target dataset. This is implemented using separate encoders for c and s and a shared decoder, together with constraints that enforce the salient representation to be inactive for background samples and to distinguish between background and target samples.

In medical imaging, this corresponds to separating normal anatomical variation from pathology-specific deviations by treating healthy images as background and pathology-specific images as target. Under this formulation, a given pathology can be isolated by constructing a dataset in which it is present only in the target set and absent from the background.

However, the SepVAE architecture isolates one factor at a time and does not provide a mechanism for representing multiple factors simultaneously within a single partitioned latent representation. In our setting, the objective is to learn a representation in which multiple factors are explicitly separated and can be composed within a single model. We therefore extend the structural partitioning principle of SepVAE from a binary commonsalient decomposition to a multi-factor latent structure in which each factor is assigned a dedicated latent component. Structural partitioning alone does not guarantee separation. When factors are correlated, their representations can leak across latent components [Scaramuzzino *et al.* 2023]. We address this by incorporating weak supervision with gradient reversal to suppress cross-factor predictability and ensure that each latent component encodes only its designated factor. In particular, this prevents distinct pathologies, such as cardiomegaly and pleural effusion, from being jointly encoded within the same latent partition.

2.2 Diffusion Models

Diffusion models [Sohl-Dickstein *et al.* 2015] are latent-variable generative models of the form $p_\theta(x_0) = \int p_\theta(x_{0:T})\,\mathrm{d}x_{1:T}$, where $x_{1:T}$ are latents of the same dimensionality as the observed data x_0. Following Ho *et al.* [2020], the model consists of a fixed *forward diffusion process* that gradually adds noise to data, and a learned *reverse process* that aims to invert this corruption.

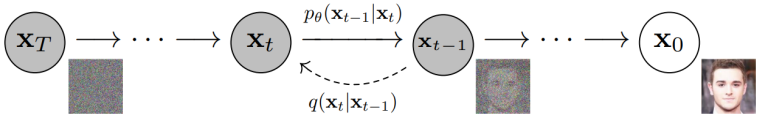


Figure 2.3: Directed graphical model structure of DDPMs [Ho *et al.* 2020].

The joint distribution of all variables is written as a Markov chain:

$$p_\theta(x_{0:T}) = p(x_T) \prod_{t=1}^{T} p_\theta(x_{t-1} \mid x_t), \qquad p(x_T) = \mathcal{N}(0, I), \tag{2.2}$$

where $p_\theta(x_{t-1} \mid x_t)$ is the reverse process. The forward diffusion process is defined as a separate Markov chain:

$$q(x_{1:T} \mid x_0) = \prod_{t=1}^{T} q(x_t \mid x_{t-1}), \qquad q(x_t \mid x_{t-1}) = \mathcal{N}\Big(\sqrt{1-\beta_t}\, x_{t-1},\ \beta_t I\Big), \tag{2.3}$$

where $\{\beta_t\}_{t=1}^{T}$ is the variance schedule that *can* be learned by reparameterization, or fixed to predetermined constants (e.g., a linear schedule from 10^{-4} to 10^{-2}) such that β_t typically increases slowly with t. This ensures that the forward chain remains analytically tractable and preserves the same functional form across timesteps while gradually driving x_T toward an isotropic Gaussian.

Forward Diffusion Process. The forward diffusion process is a fixed Markov chain that gradually corrupts a clean sample x_0 by adding Gaussian noise. Following Equation 2.3, each transition has the form

$$q(x_t \mid x_{t-1}) = \mathcal{N}\Big(\sqrt{1-\beta_t}\, x_{t-1},\ \beta_t I\Big). \tag{2.4}$$

Because these transitions are linear Gaussian, the marginal distribution at any timestep is available in closed form:

$$q(x_t \mid x_0) = \mathcal{N}\big(\sqrt{\bar{\alpha}_t}\, x_0,\ (1-\bar{\alpha}_t) I\big), \quad \text{where} \quad \alpha_t := 1-\beta_t, \qquad \bar{\alpha}_t := \prod_{s=1}^{t} \alpha_s. \tag{2.5}$$

Equivalently, x_t can be written through the reparameterization

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, I). \tag{2.6}$$

This makes explicit that the state at timestep t is a linear combination of the original sample and Gaussian noise. As t increases, the signal coefficient $\sqrt{\bar{\alpha}_t}$ decreases, so the forward process progressively destroys structure and drives x_t toward an isotropic Gaussian distribution.

Reverse Denoising Process. The generative task is to invert the forward corruption process. To achieve this, Ho *et al.* [2020] learn a reverse Markov chain whose transitions are Gaussian:

$$p_\theta(x_{t-1} \mid x_t) = \mathcal{N}\big(\mu_\theta(x_t, t),\ \Sigma_\theta(x_t, t)\big). \tag{2.7}$$

Here, θ denotes the parameters of a single neural network shared across timesteps, and the timestep t is provided to the network through a timestep embedding.

Figure 2.4 gives an overall picture of the diffusion process, showing that the forward and reverse chains pass through the same sequence of latent states in opposite directions.

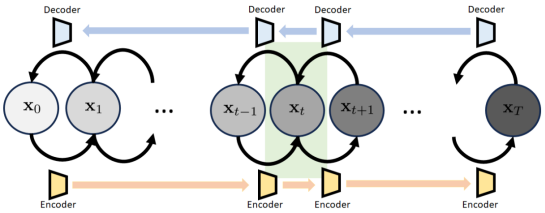


Figure 2.4: Kingma *et al.* [2021] illustrate a variational diffusion model in which the lower path represents the forward process that gradually corrupts x_0 into x_T, while the upper path represents the reverse process that denoises step by step. The intermediate states $x_1, \ldots, x_{T-1}$ are latent variables of the same dimensionality as the data [Chan 2024].

Because the forward process is Gaussian and fixed, it induces an exact conditional distribution over the previous state when both the current noisy state x_t and the original clean sample x_0 are known:

$$q(x_{t-1} \mid x_t, x_0) = \mathcal{N}\big(\tilde{\mu}_t(x_t, x_0),\ \tilde{\beta}_t I\big). \tag{2.8}$$

Its mean and variance are

$$\tilde{\mu}_t(x_t, x_0) = \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t}\, x_0 + \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}\, x_t, \tag{2.9}$$

$$\tilde{\beta}_t = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\, \beta_t. \tag{2.10}$$

It is important to distinguish the reverse conditional in Equation 2.8 from the forward transition in Equation 2.4. Since the forward process is Markovian,

$$q(x_t \mid x_{t-1}, x_0) = q(x_t \mid x_{t-1}), \tag{2.11}$$

so the next noisy state depends only on the previous noisy state. In contrast, the exact reverse conditional in Equation 2.8 depends on both x_t and x_0. During training, x_0 is known, so this conditional can be computed exactly. At sampling time, however, x_0 is unknown, so the model learns the practical approximation $p_\theta(x_{t-1} \mid x_t)$ in Equation 2.7.

Figure 2.5 highlights this local asymmetry. The forward transition is fixed and Gaussian, while the reverse transition must be learned.

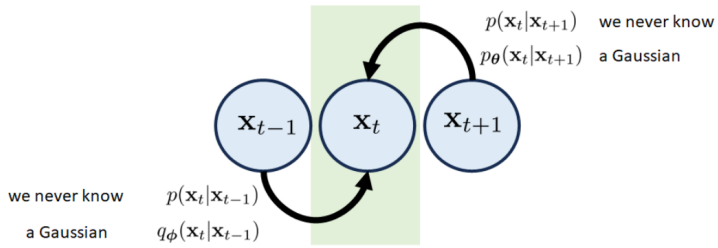


Figure 2.5: Local transition block in a diffusion model. The blue circles denote neighboring latent states in the diffusion chain: x_{t-1} is a less noisy state, x_t is the current state, and x_{t+1} is a more noisy state. The forward process moves from left to right, gradually adding noise, while the reverse process moves from right to left, progressively denoising. The forward transition $q(x_t \mid x_{t-1})$ is the fixed Gaussian noising step that maps a cleaner state to a noisier one. The learned reverse transition $p_\theta(x_{t-1} \mid x_t)$ maps a noisy state to a less noisy one and is used at sampling time. The quantity $p(x_{t-1} \mid x_t)$ denotes the ideal but generally intractable reverse transition, while the exact reverse conditional $q(x_{t-1} \mid x_t, x_0)$ becomes tractable during training because the original clean sample x_0 is known. Thus, the forward process uses the fixed Gaussian transition $q(x_t \mid x_{t-1})$, while the reverse process learns $p_\theta(x_{t-1} \mid x_t)$. The exact reverse conditional $q(x_{t-1} \mid x_t, x_0)$ is available during training because x_0 is known, but not during sampling [Chan 2024].

Training Objective. Training seeks to maximize the likelihood of the data under the learned reverse process. Since the latent variables $x_{1:T}$ cannot be marginalized exactly, training instead optimizes a standard variational lower bound:

$$\mathbb{E}_q[-\log p_\theta(x_0)] \;\le\; \mathcal{L}(\theta) = \mathbb{E}_q\left[-\log \frac{p_\theta(x_{0:T})}{q(x_{1:T} \mid x_0)}\right]. \tag{2.12}$$

This can be rewritten as the familiar decomposition

$$\mathcal{L}(\theta) = \mathbb{E}_q\Big[\underbrace{D_{\mathrm{KL}}\big(q(x_T \mid x_0)\,\|\,p(x_T)\big)}_{L_T} + \sum_{t=2}^{T}\underbrace{D_{\mathrm{KL}}\big(q(x_{t-1} \mid x_t, x_0)\;\|\;p_\theta(x_{t-1} \mid x_t)\big)}_{L_{t-1}} - \underbrace{\log p_\theta(x_0 \mid x_1)}_{L_0}\Big]. \tag{2.13}$$

The term L_T matches the terminal forward distribution to the prior. The intermediate terms L_{t-1} compare the learned reverse transition in Equation 2.7 with the exact conditional in Equation 2.8. The final term L_0 plays the role of a reconstruction term for x_0.

Simplified Noise-Prediction Objective. The variational bound in Equation 2.13 decomposes as $L_{\text{VLB}} = L_T + \sum_{t=2}^{T} L_{t-1} + L_0$. The terms L_T and L_0 do not depend on how the learned reverse mean is parameterized, so the key modeling choice appears in the intermediate terms L_{t-1}.

Since both $q(x_{t-1} \mid x_t, x_0)$ and $p_\theta(x_{t-1} \mid x_t)$ are Gaussian, and the reverse-process variance is fixed, each intermediate term reduces to matching the learned mean $\mu_\theta(x_t, t)$ to the true posterior mean $\tilde{\mu}_t(x_t, x_0)$:

$$L_{t-1} = \mathbb{E}_q \left[\frac{1}{2\sigma_t^2} \left\| \tilde{\mu}_t(x_t, x_0) - \mu_\theta(x_t, t) \right\|^2 \right] + C_t, \tag{2.14}$$

where C_t does not depend on θ.

Using the forward reparameterization in Equation 2.6, the true posterior mean in Equation 2.9 can be rewritten as

$$\tilde{\mu}_t(x_t, x_0) = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \, \varepsilon \right). \tag{2.15}$$

This suggests a convenient parameterization of the learned reverse mean. Instead of predicting $\mu_\theta(x_t, t)$ directly, the network predicts the noise term $\varepsilon_\theta(x_t, t)$, and the model mean is then defined by

$$\mu_\theta(x_t, t) = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \, \varepsilon_\theta(x_t, t) \right). \tag{2.16}$$

Thus, the network is trained to predict noise, while this prediction implicitly determines the learned reverse mean.

Substituting Equations 2.15 and 2.16 into Equation 2.14 gives

$$L_{t-1} = \mathbb{E}_{x_0, \varepsilon} \left[\frac{(1 - \alpha_t)^2}{2\sigma_t^2 \, \alpha_t (1 - \bar{\alpha}_t)} \left\| \varepsilon - \varepsilon_\theta(x_t, t) \right\|^2 \right] + C_t, \tag{2.17}$$

so each intermediate KL term becomes a weighted mean-squared error between the true noise ε and the noise predicted by the network.

Dropping the timestep-dependent weight yields the simplified objective used in practice:

$$L_{\text{simple}}(\theta) = \mathbb{E}_{t, x_0, \varepsilon} \left[\|\varepsilon - \varepsilon_\theta(x_t, t)\|^2 \right], \qquad x_t = \sqrt{\bar{\alpha}_t} \, x_0 + \sqrt{1 - \bar{\alpha}_t} \, \varepsilon. \tag{2.18}$$

This is a reweighted form of the ELBO, but is simpler to optimize and is commonly used because it gives stable training and high-quality samples.

Training and Sampling Algorithms. Training and sampling use the same network in different ways. During training, the model does not execute the full reverse chain. Instead, it draws a clean sample x_0, samples a timestep t uniformly from $\{1, \ldots, T\}$, samples noise $\varepsilon \sim \mathcal{N}(0, I)$, and constructs x_t directly using Equation 2.6. The network then predicts $\varepsilon_\theta(x_t, t)$, and a gradient step is taken on the single-timestep loss in Equation 2.18. Thus, the gradient is computed from the loss at the sampled noise level t; training does not backpropagate through the full reverse trajectory.

This still trains one model to handle all timesteps because the same network is shared across the entire noise schedule and receives t as an input. By sampling different values of t during training, the model learns to denoise across all noise levels.

The full sequential reverse chain appears only during sampling. Starting from $x_T \sim \mathcal{N}(0, I)$, the model repeatedly applies the reverse update

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \, \varepsilon_\theta(x_t, t) \right) + \sigma_t z, \qquad z \sim \mathcal{N}(0, I), \tag{2.19}$$

for $t = T, T-1, \ldots, 1$, where σ_t^2 is fixed by the chosen reverse-process parameterization. This iterative reverse process gradually transforms noise into a structured sample x_0.

2.2.1 Score-Based Formulation

Song et al. [Song *et al.* 2021] showed that DDPMs can be understood in continuous time through stochastic differential equations. In this view, discrete-time diffusion models and score-based generative models are two parameterizations of the same underlying denoising process.

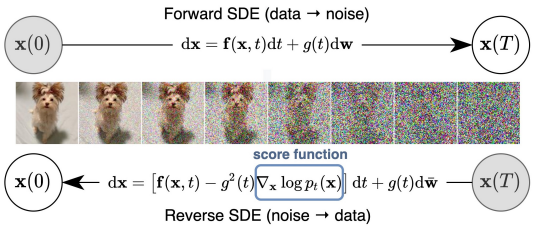


Figure 2.6: **Forward and reverse SDEs.** In continuous time, the forward process corrupts data with drift and diffusion terms, while the reverse process uses the score function to steer noisy samples back toward the data manifold [Song *et al.* 2021].

From the Markov Chain to the Continuous Limit. The discrete forward chain can be interpreted as a discretized version of a continuous corruption process. In continuous time, the variance-preserving formulation is described by

$$\mathrm{d}x = -\tfrac{1}{2}\beta(t)\, x\, \mathrm{d}t + \sqrt{\beta(t)}\, \mathrm{d}W(t), \tag{2.20}$$

where $W(t)$ is a Wiener process. More generally, the forward corruption process is written as

$$\mathrm{d}x = f(x, t)\, \mathrm{d}t + g(t)\, \mathrm{d}w, \tag{2.21}$$

with drift $f(x, t)$ and diffusion strength $g(t)$.

Reverse SDE. The corresponding reverse-time SDE has the form

$$\mathrm{d}x = \left[f(x, t) - g(t)^2 \nabla_x \log p_t(x)\right] \mathrm{d}t + g(t)\, \mathrm{d}\bar{w}, \tag{2.22}$$

where $\nabla_x \log p_t(x)$ is the *score function*, namely the gradient of the log-density at noise level t. Intuitively, the score points in the direction that makes the current sample more likely under the time-dependent data distribution. The reverse SDE is the continuous-time counterpart of the DDPM reverse chain where reverse denoising is driven by a vector field that points back toward the data manifold.

In the DDPM parameterization, the model outputs $\varepsilon_\theta(x_t, t)$, an estimate of the Gaussian noise used to corrupt x_0 into x_t. Because x_t is a linear combination of signal and noise, this noise estimate can be converted directly into a score estimate:

$$s_\theta(x_t, t) = -\frac{\varepsilon_\theta(x_t, t)}{\sqrt{1 - \bar{\alpha}_t}}. \tag{2.23}$$

DDPMs can therefore be viewed as score-based generative models written in a different parameterization.

2.2.2 Conditional Generation and Classifier-Free Guidance

This section distinguishes between **unconditional** and **conditional** diffusion models by examining their objectives, formulations, and applications.

While the above formulation defines the foundation of *unconditional diffusion models*, practical applications often demand controlled or semantically guided generation such as synthesizing specific classes, attributes, or text-aligned imagery. This introduces the need for *conditional diffusion models*, where additional information y (e.g., labels, text embeddings, or medical attributes) guides the reverse process. Recent extensions, such as *classifier* [Dhariwal and Nichol 2021] and *classifier-free guidance* [Ho and Salimans 2022], further refine this conditioning by explicitly or implicitly modulating the score function to balance fidelity and diversity during sampling.

Unconditional Diffusion Models. An *unconditional diffusion model* learns to generate samples from the marginal data distribution $p(x)$ without using auxiliary information or labels. During training, the model observes only samples x_0 drawn from $p_{\text{data}}(x)$ and is optimized to predict the noise added at each diffusion step:

$$\mathcal{L}_{\text{uncond}} = \mathbb{E}_{x_0,t,\varepsilon}\left[\|\varepsilon - \varepsilon_\theta(x_t, t)\|^2\right]. \tag{2.24}$$

The training signal is generated from the data itself rather than supplied by external annotation, so the objective is self-supervised. At sampling time, the model starts from Gaussian noise $x_T \sim \mathcal{N}(0, I)$ and denoises it step by step to produce a realistic sample. In this sense, unconditional diffusion models learn $p(x)$ directly, without any conditioning variable.

Conditional Diffusion Models. A *conditional diffusion model* extends the unconditional setting by modeling $p(x \mid y)$, where y provides additional information such as a class label, text prompt, or spatial map. The forward noising process is unchanged. Conditioning enters only through the reverse model, which is now written as

$$p_\theta(x_{t-1} \mid x_t, y).$$

In the DDPM parameterization, this is implemented by conditioning the noise predictor $\varepsilon_\theta(x_t, t, y)$. Training therefore uses the objective

$$\mathcal{L}_{\text{cond}} = \mathbb{E}_{x_0,t,\varepsilon,y}\left[\|\varepsilon - \varepsilon_\theta(x_t, t, y)\|^2\right]. \tag{2.25}$$

By Equation 2.23, this is equivalent to learning the conditional score of the noisy data distribution. At sampling time, the condition y steers the denoising trajectory toward samples consistent with the desired class, prompt, or attribute. In this way, conditional diffusion turns unconditional generation into controllable generation.

Classifier Guidance. Classifier guidance [Dhariwal and Nichol 2021] provides a post-hoc way to steer an unconditional diffusion model toward a target condition y. The key observation is that a classifier gives access to $p_\phi(y \mid x_t)$, and therefore to the gradient

$$\nabla_{x_t} \log p_\phi(y \mid x_t).$$

At timestep t, the goal is to obtain the conditional score of the noisy-state distribution $p_t(x_t \mid y)$. Using the product rule for the joint distribution,

$$p_t(x_t, y) = p_\phi(y \mid x_t)\, p_t(x_t) = p_t(x_t \mid y)\, p(y).$$

Rearranging the terms yields

$$p_t(x_t \mid y) = \frac{p_\phi(y \mid x_t)\, p_t(x_t)}{p(y)}.$$

Therefore,

$$\log p_t(x_t \mid y) = \log p_\phi(y \mid x_t) + \log p_t(x_t) - \log p(y),$$

and differentiating with respect to x_t yields

$$\nabla_{x_t} \log p_t(x_t \mid y) = \nabla_{x_t} \log p_t(x_t) + \nabla_{x_t} \log p_\phi(y \mid x_t),$$

since $p(y)$ does not depend on x_t.The conditional score can therefore be written as the sum of the unconditional score and a classifier-derived conditioning term. In practice, Dhariwal and Nichol [2021] scale this conditioning term by a guidance weight $\gamma \geq 0$:

$$\tilde{s}(x_t, t, y) = s_\theta(x_t, t) + \gamma\, \nabla_{x_t} \log p_\phi(y \mid x_t). \tag{2.26}$$

Larger values of γ strengthen adherence to the target condition, often at the cost of reduced sample diversity.

Classifier-Free Guidance. Classifier-free guidance (CFG) [Ho and Salimans 2022] achieves a similar effect without training a separate classifier. A single diffusion model is trained with condition dropout so that it learns both the conditional and unconditional predictors

$$\varepsilon_\theta(x_t, t, y) \qquad \text{and} \qquad \varepsilon_\theta(x_t, t, \emptyset),$$

where $\emptyset$ denotes the null condition. At sampling time, these two predictions are combined as

$$\tilde{\varepsilon}_\theta(x_t, t, y) = (1 + w)\, \varepsilon_\theta(x_t, t, y) - w\, \varepsilon_\theta(x_t, t, \emptyset), \tag{2.27}$$

with guidance weight $w \geq 0$. When $w = 0$, this reduces to ordinary conditional generation. Larger values of w strengthen the conditional signal while reducing diversity. This formulation has become the standard conditioning mechanism in modern diffusion models. In the proposed study, it plays two concrete roles. This is relevant throughout the proposal because classifier-free guidance provides the basic conditioning mechanism used for standard prompting and for composition. In Phase 1, it underlies both the single-prompt baseline and the compositional operator of Liu *et al.* [2022]. In Phases 2 and 3, the same conditional prediction framework is used in the controlled synthetic and clinical composition settings.

Stable Diffusion. Stable Diffusion [Rombach *et al.* 2022] is a *latent diffusion model*, meaning that diffusion is performed in a compressed latent space rather than directly in pixel space. An image x is first mapped by a variational autoencoder into a latent representation z, and the forward and reverse diffusion processes are learned over this lower-dimensional latent variable. T his substantially reduces the computational cost of training and sampling while retaining the ability to decode high-resolution images. For text-to-image generation, a text encoder produces prompt embeddings that are injected into the U-Net denoiser through cross-attention [Vaswani *et al.* 2017], allowing the denoising trajectory to remain aligned with the prompt semantics. During training, the model learns to predict the noise added to latent variables under text conditioning. During sampling, it starts from Gaussian latent noise, iteratively denoises using the text-conditioned predictor with classifier-free guidance, and then decodes the final latent through the VAE decoder to obtain an image.

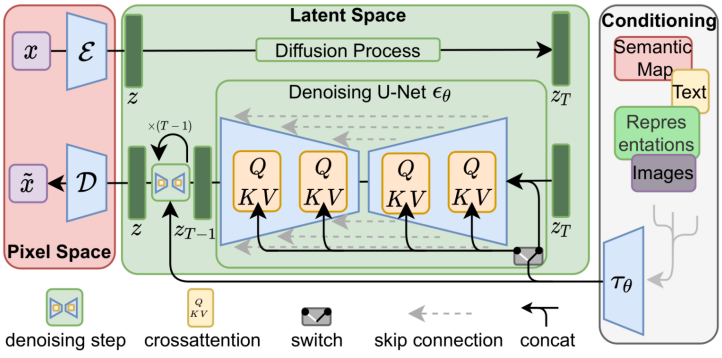


Figure 2.7: Stable Diffusion as a latent text-to-image diffusion model. A VAE maps images to a compressed latent space, a text encoder provides prompt embeddings, and a U-Net denoises latent noise under text conditioning before the VAE decoder reconstructs the final image. Operating in latent space reduces computational cost while preserving high-quality generation.

2.2.3 Vision-Language Models and Prompt Inversion

CLIP. CLIP [Radford *et al.* 2019] learns a shared embedding space for images and text using a contrastive objective over paired image-caption data. Its central contribution is to align visual and textual representations so that matching image-text pairs have high similarity while mismatched pairs are pushed apart. This shared space is useful for text-to-image systems because it provides a generic notion of semantic correspondence between prompts and images. In the context of the proposed study, CLIP is relevant as a representation for prompt-image matching and as a building block for inversion methods that map generated images back toward text-aligned embeddings.

BLIP. BLIP [li2] is a vision-language model for both image understanding and language generation. It combines image and text encoders with multimodal pre-training to support tasks such as captioning, retrieval, and visual question answering. Unlike contrastive models such as CLIP, which mainly return cross-modal similarity scores, BLIP can generate text conditioned on an image. This makes it useful when the evaluation pipeline requires a textual description of generated outputs rather than only an image-text matching score.

BLIP-VQA denotes the use of BLIP-style vision-language models for visual question answering, where the model is conditioned on both an image and a natural-language question and returns a textual answer. In this setting, the model can be asked whether particular entities, attributes, or relations are present in a generated image, making it useful for targeted evaluation of compositional correctness. For the proposed study, BLIP-VQA is relevant as a possible tool for querying whether a generated sample actually contains the intended concepts, rather than relying only on global similarity measures or free-form captions.

Prompt Inversion. In the CLIP [Radford *et al.* 2019] text encoder used by Stable Diffusion [Rombach *et al.* 2022; Podell *et al.* 2023], the End-of-Sentence (EOS) token has an important role beyond simply marking the end of a prompt. Because CLIP uses a causal attention mask, each token can attend only to the tokens that come before it. As the final token in the sequence, the EOS position is therefore the only one whose hidden state has access to the entire prompt. When CLIP forms a single text representation for image-text alignment, it uses this EOS hidden state rather than averaging all token embeddings. In this sense, the EOS representation acts as a contextualized summary of the whole prompt.

Stable Diffusion Image-to-Prompt Conversion (SD-IPC) [Ding *et al.* 2023a] builds on this property to convert a reference image into a Stable Diffusion text-conditioning signal with little additional training. The method relies on two observations: first, that Stable Diffusion generation is strongly influenced by this EOS-based text representation; and second, that CLIP training aligns the projected EOS text embedding with the corresponding visual embedding. Using this relationship, the authors derive a closed-form pseudo-inverse of the CLIP text projection matrix to map a CLIP image embedding into the prompt-side conditioning space. This converted embedding is then repeated to form a pseudo-prompt sequence. In the context of this study, SD-IPC is useful because it provides a concrete way to test whether a target image can be recovered from Stable Diffusion's prompt-conditioning space.

2.2.4 Compositional Diffusion Models

Single text-to-image diffusion models often fail when a prompt requires several factors to be combined correctly. Recent analyses [Huang *et al.* 2025; Du and Kaelbling 2024; Bradley *et al.* 2025] show consistent errors in attribute binding, spatial relations, and numeracy, even when the image remains visually plausible. These failures arise because the models largely learn correlations in joint image-text data rather than an explicit factorized structure over objects, attributes, and relations [Huang *et al.* 2025]. In the proposed study, we refer to the conventional case in which a single natural-language prompt specifies all desired factors and the model attempts to realize them jointly as *semantic composition*.

The next three definitions review the main formulations of composition in this literature: the *simple product*, *Bayes composition*, and *projective composition*. The first two define composition by combining densities directly, while the third defines it through what the composed result should preserve.

Simple Product

The first definition is the *simple product*. Given two distributions p_1 and p_2 over $\mathbb{R}^n$, it defines the composition as

$$\hat{p}(x) \propto p_1(x)p_2(x).$$

This is a familiar way to combine distributions, but it has an important limitation: the composed distribution cannot be truly out-of-distribution with respect to p_1 or p_2. If either $p_1(x) = 0$ or $p_2(x) = 0$, then $\hat{p}(x) = 0$ as well. As Bradley *et al.* [2025] show, this becomes a problem when the desired composition lies outside the support of both source distributions. In their CLEVR example, if each source distribution contains only one object, then any image with *both* objects has zero probability under both p_1 and p_2, so the simple product also assigns it zero probability.

Bayes Composition

A second prior definition is *Bayes composition*. Rather than multiplying arbitrary densities, Bayes composition treats the component distributions as conditionals of a common underlying data distribution. [Liu *et al.* 2022; Du *et al.* 2023; Du and Kaelbling 2024; Du *et al.* 2020]. If c_1 and c_2 denote two conditions of interest, the desired composition is defined as

$$\hat{p}(x) := p(x \mid c_1, c_2).$$

Assuming c_1 and c_2 are conditionally independent given x, Bayes' rule gives

$$\hat{p}(x) = p(x \mid c_1, c_2) \propto \frac{p(x \mid c_1)\, p(x \mid c_2)}{p(x)}. \tag{2.28}$$

This formulation does not define a truly out-of-distribution composition [Bradley *et al.* 2025], because it works when the desired conjunction already corresponds to a meaningful conditional under the training distribution, but cannot in general create a new conjunction that lies outside that support.

For diffusion models, this composition is implemented at the level of the score. Taking $\nabla_x \log$ of Equation 2.28 gives

$$\nabla_x \log \hat{p}(x \mid c_1, c_2) = \nabla_x \log p(x \mid c_1) + \nabla_x \log p(x \mid c_2) - \nabla_x \log p(x). \tag{2.29}$$

Thus, the composed score is the sum of the two conditional scores minus the unconditional score.

Using the score estimate in Equation 2.23, the score composition in Equation 2.29 can be written directly in terms of noise predictions. Let $\varepsilon_\theta(x_t, t, \varnothing)$ denote the unconditional prediction and $\varepsilon_\theta(x_t, t, c_i)$ the prediction conditioned on concept c_i. For a single condition, classifier-free guidance takes the form

$$\varepsilon_\theta(x_t, t, \varnothing) + w_i\big[\varepsilon_\theta(x_t, t, c_i) - \varepsilon_\theta(x_t, t, \varnothing)\big].$$

Applying this construction to both concepts yields the practical AND operator used by Liu *et al.* [2022]:

$$\tilde{\varepsilon}_{\text{AND}} = \varepsilon_\theta(x_t, t, \varnothing) + w_1\big[\varepsilon_\theta(x_t, t, c_1) - \varepsilon_\theta(x_t, t, \varnothing)\big] + w_2\big[\varepsilon_\theta(x_t, t, c_2) - \varepsilon_\theta(x_t, t, \varnothing)\big]. \tag{2.30}$$

When $w_1 = w_2 = 1$, this reduces to

$$\varepsilon_\theta(x_t, t, c_1) + \varepsilon_\theta(x_t, t, c_2) - \varepsilon_\theta(x_t, t, \varnothing),$$

which is exactly the Bayes composition written in noise-prediction form.

Projective Composition

Bradley *et al.* [2025] argue that a useful definition of composition must answer two separate questions: what the composed distribution should preserve, and when a practical compositional method can realize it. Their notion of *projective composition* addresses the first question. Under this view, the composed distribution is judged by the part each source is meant to contribute. For example, if one source determines style and another content, then a composition is correct if it preserves the style of the first and the content of the second, even when neither source distribution contains that full combination. This is closely aligned with Du and Kaelbling [2024], where composition is understood as building a new distribution from simpler components that each contribute part of the final structure.

This matters because it allows composition that is out of distribution with respect to the source distributions being combined. The final sample can combine properties from different sources without needing to belong to any one source distribution as a whole. As Figure 2.8 illustrates, the composed distribution can differ from both source distributions in the original sample space while still matching each source after applying the corresponding projection. That is precisely what the simple product cannot express, since it is confined to overlapping support, and what Bayes composition cannot justify when the desired conjunction is not already supported by some underlying joint distribution.

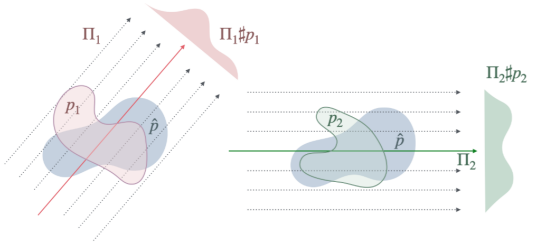


Figure 2.8: Illustration of projective composition. The composed distribution $\hat{p}$ need not coincide with either source distribution p_1 or p_2 in the original sample space. Instead, correctness is defined after applying source-specific projections: under Π_1, the composed distribution matches the aspect preserved from p_1, and under Π_2, it matches the aspect preserved from p_2. The figure therefore shows how a composition can be out of distribution with respect to the sources as whole distributions while still preserving the designated contribution of each source.

The key implication is representational. Standard disentanglement is not enough if concepts still interfere when one is changed. What composition requires is a stronger separation in which each concept acts on its own factor without perturbing the others. Bradley *et al.* [2025] further show that this separation need not appear in pixel space itself. If it exists in some invertible latent space, then the same composition operator yields a correct projective composition with respect to those latent space factors. This is a statement about when the target composition is well defined, not a guarantee that reverse-diffusion sampling will always recover it. For the proposed study, the lesson is that compositional success depends on whether the learned representation allows the concepts to be preserved independently.

Chapter 3

Related Work

3.1 Compositional Generative Models

Early approaches decomposed data into disentangled factors of variation, learning representations in a continuous latent space where each dimension corresponds to a semantic factor [Higgins *et al.* 2017; Vedantam *et al.* 2018]. Such factorized representations allowed interpretable control but were limited by their fixed latent dimensionality, which restricted the ability to dynamically introduce or recombine new factors. Complementary efforts in spatial decomposition treated images as structured collections of independent object slots [van Steenkiste *et al.* 2019; Greff *et al.* 2019]. These methods excelled in representing multi-object scenes but often failed to model the interactions or dependencies between objects, which are crucial for understanding relational structure.

Energy-Based Models (EBMs) introduced a unifying probabilistic formulation for compositionality. Rather than relying on explicit latent variables, EBMs define an energy function $E_\theta(\mathbf{x})$ that maps an input $\mathbf{x}$ to a scalar energy value, with lower energies corresponding to more likely configurations. Following the product-of-experts formulation introduced by Hinton [1999], independent energy functions can be combined to represent conjunctions of concepts:

$$p(\mathbf{x} \mid c_1, c_2, \ldots, c_n) \propto \exp\Big(-\sum_i E_\theta(\mathbf{x} \mid c_i) \Big).$$

This additive composition provides a mathematically elegant mechanism for representing logical operations such as conjunction (AND), disjunction (OR), and negation (NOT) through energy summation or inversion. Later works such as Shazeer *et al.* [2017]; Greff *et al.* [2019] formalized the notion that multiple energy landscapes can be recursively combined to model increasingly complex visual structures, while Du *et al.* [2020] demonstrated how Markov Chain Monte Carlo (MCMC) sampling on the composed energy surface enables the generation of images consistent with multiple attributes simultaneously.

Diffusion Models as Energy-Based Systems. A central insight underpinning recent advances is that diffusion models can be interpreted as a class of energy-based models [Du *et al.* 2023]. The score function $\nabla_{\mathbf{x}} \log p(\mathbf{x})$ of a diffusion model defines a gradient field over data space, which, like an energy landscape, governs the movement of samples toward regions of high likelihood.

Recent work has sought to operationalize compositionality within the framework of diffusion models. Liu *et al.* [2022] propose a structured approach in which separately trained diffusion models, each modeling a distinct component of an image, are algebraically combined at inference time. By viewing diffusion models as energy-based systems, they define compositional operators, such as logical conjunction (AND) and negation (NOT), that allow for flexible test-time control without additional training. This formulation enables zero-shot compositional generalization, allowing the model to generate coherent images that represent novel concept combinations far beyond its training distribution.

More recently, Skreta *et al.* [2024] proposed a framework for combining pretrained diffusion models under the principle of *superposition*. Their algorithm, SuperDiff, leverages a scalable Itô density estimator for the diffusion stochastic differential equation (SDE), allowing for adaptive reweighting of log-likelihoods

during composition. This method enables logical operations such as conjunction and disjunction between pretrained models directly during inference, without retraining or fine-tuning. The authors further demonstrate that the superposition mechanism permits fine-grained temperature control, balancing diversity and coherence in composed samples.

3.2 Synthetic Chest X-ray Generation for Medical Imaging

Recent advances in generative modeling have made synthetic image generation an important tool in medical imaging, especially for chest radiography. This is particularly useful where data is scarce, sensitive, or imbalanced across disease classes. Publicly available chest X-ray datasets such as NIH ChestX-ray [Wang *et al.* 2017] and CheXpert [Irvin *et al.* 2019] remain limited by class imbalance, label quality, scale, and privacy constraints. Generative models offer a way to address these limitations by producing anatomically plausible, pathology-aware images that can supplement training data for classification, segmentation, and detection tasks. Synthetic images may also help reduce ethical and legal concerns by replacing identifiable patient data with artificial data that preserve diagnostic relevance [Zhou *et al.* 2025].

Early Generative Approaches Using GANs. Early research on synthetic medical imaging predominantly relied on generative adversarial networks (GANs), which model the data distribution via adversarial training between a generator and discriminator. Frid-Adar *et al.* [2018] and Shin *et al.* [2018] demonstrated that GAN-based augmentation improves convolutional neural network (CNN) performance for lesion classification and anonymization tasks. However, despite their success in general image synthesis, GANs often struggle to maintain fine-grained anatomical and pathological consistency in medical images. The inherent instability of adversarial training and the difficulty of enforcing pixel-level medical realism limit the clinical reliability of GAN-generated data. Consequently, research has shifted toward more stable, probabilistically grounded diffusion-based generative models.

Diffusion Models for Chest X-ray Synthesis. Diffusion models have recently become the dominant approach for high-quality synthetic image generation due to their stability and capacity to model complex data distributions. Mahaulpatha *et al.* [2024] introduced a Denoising Diffusion Probabilistic Model (DDPM) for X-ray synthesis, leveraging iterative noisy-to-clean transformations to reconstruct diverse, realistic radiographs. Unlike GANs, DDPMs explicitly learn the score function of the data distribution, allowing them to generate anatomically consistent samples without adversarial objectives. Their probabilistic nature provides smoother convergence and better control over generation trajectories, making them particularly suitable for medical domains where subtle textural and structural details are critical [Meng *et al.* 2022].

More recent work has transitioned from pixel-space diffusion to latent-space diffusion for improved efficiency and control. Prakash *et al.* [2025] proposed a latent diffusion framework for generating synthetic chest X-rays conditioned on segmentation masks and textual prompts. By guiding the generative process through anatomical or semantic conditions, their method produced pathology-controllable radiographs that enhance the diversity of training data while preserving clinical realism. Their study also integrated radiologist feedback to iteratively refine the generation process a "human-in-the-loop" strategy that bridges generative modeling and expert validation. Such approaches directly address the issue of limited publicly available annotated data by enabling targeted synthesis of underrepresented disease cases.

Beyond unconditional synthesis, controlling the generation of anatomically and pathologically distinct regions has become a central research focus. Hashmi *et al.* [2024] introduced *XReal*, a pathology- and anatomy-aware diffusion model that introduces spatial control via dedicated anatomy and pathology controllers. By adapting pretrained text-to-image diffusion models, *XReal* enables explicit manipulation of anatomical structures (e.g., lungs, ribs) and pathological regions (e.g., lesions, infiltrates) without requiring full model retraining.

To our knowledge, compositional diffusion has not yet been studied in medical imaging as a method for generating out-of-distribution samples that represent clinically meaningful co-morbid presentations. The proposed study addresses this gap by combining disease-specific diffusion models at inference time to synthesize chest X-rays that reflect multiple pathologies simultaneously, even when such co-morbid cases are scarce or absent in the training data. The contribution is a generative framework for rare co-morbid synthesis in medical imaging.

Chapter 4

Research Objectives

4.1 Introduction

The previous chapter established the motivation for the proposed research. This chapter defines the research goals aims of the proposed study.

4.2 Research Goals

The primary research objectives are as follows:

1. **Characterise the Composability Gap.** Measure how severely logical composition diverges from semantic composition, and determine whether this gap varies systematically across concept pairs grouped by co-occurrence support, feature-space disentanglement, or semantic conflict, establishing it as a predictable property of concept pairs rather than a set of isolated failure cases.
2. **Determine the Cause of the Gap.** Identify whether the gap stems from how concepts are represented in the latent space or from the compositional method used at inference time, thereby locating the main bottleneck to reliable compositional generation.
3. **Demonstrate Practical Utility in Medical Imaging.** Test whether compositional generative models can produce plausible rare co-morbid presentations at inference time, without joint training examples, demonstrating that the earlier insights are practically useful in low-resource clinical settings.

4.3 Research Hypotheses

The main hypotheses of this proposal are:

- **H1 (The Composability Gap):** Logical composition via score addition produces a measurable composability gap relative to semantic composition, emerging early in inference and following a principled ordering: smallest for co-occurring pairs, small-to-moderate for approximately disentangled (e.g style-content pairs) unseen pairs, large for entangled pairs, and largest for semantically competing pairs. The largest-gap outcomes should be unreachable via natural-language prompting alone, confirming that logical composition steers generation along paths no single prompt can reproduce.
- **H2 (The Representation Hypothesis):** The composability gap is primarily a representation problem. Holding the compositional method fixed, models with more clearly separated concept representations should exhibit a smaller composability gap than models with more entangled ones.
- **H3 (The Clinical Independence Sufficiency Hypothesis):** Approximate clinical independence between chest X-ray pathologies is sufficient for logical composition to generate plausible co-morbid outputs, including out-of-distribution pairings unseen during training, provided the model has learned sufficiently distinct pathology-specific representations.

Chapter 5

Research Method

5.1 Phase 1: Characterising the Composability Gap

As discussed in Section 2.2.4, Bayes-style score composition does not define a truly out-of-distribution joint distribution. Its success therefore depends on the model already containing a usable approximation to the desired joint configuration within its learned training distribution; when that assumption fails, score-level composition can deviate substantially from the behaviour of a model conditioned on a single semantic description of the same scene.

This leads to a central problem:

> **Logical composition and semantic composition are not equivalent, and their discrepancy depends on the structure of the underlying data distribution.**

In this phase, we empirically characterise this discrepancy as a *composability gap*, and study how it varies across different types of concept pairs. To do so, we construct controlled groups of compositions designed to span a range of regimes from approximately factorized pairs to strongly conflicting or entangled concepts. This grouping allows us to probe the structural conditions under which logical composition succeeds or fails, and to identify the mechanisms underlying this gap.

This phase tests Hypothesis 1 in three steps:

- We establish that the composability gap exists and varies systematically across concept pairs chosen to span a range of conditions, from manifold-supported co-occurrence to adversarial concept collision.
- We characterise the gap dynamically by tracking full denoising trajectories, allowing us to identify when the two generation processes diverge and how that divergence develops over time.
- We test whether the endpoint reached by logical composition can be recovered through language conditioning alone. Persistent failure here supports the claim that logical composition reaches targets that standard semantic prompting does not.

5.1.1 Concept Pair Taxonomy

The taxonomy is grounded in the conditions under which Bradley *et al.* [2025] predicts composition to succeed or fail. Rather than treating successes and failures as isolated examples, Phase 1 groups concept pairs into interpretable regimes so that H1 can be tested against a clear structural ordering.

We partition concept pairs into four groups:

1. **Co-occurring Concepts (Group 1):** These pairs frequently co-occur in the training distribution, so the model is likely to have already learned a usable joint representation. This serves as a positive control: logical and semantic composition should remain closely aligned, yielding the smallest gaps.
2. **Feature-Space Disentanglement (Group 2):** These pairs act on largely distinct feature dimensions, such as object–style combinations. They form a second positive control: even without

strong co-occurrence in the training data, composition should remain reliable when the factors are structurally compatible.

3. **Feature Overlap (Group 3):** These pairs have weak joint support and no clear feature-space separation. They are expected to produce intermediate behaviour: larger and more variable gaps than the positive controls, but not necessarily universal failure.
4. **Adversarial Collision (Group 4):** These pairs compete for the same semantic slot, object role, or spatial allocation. They represent the strongest violation of the conditions required for successful composition, and should therefore produce the largest and most systematic failures.

This taxonomy defines the pair regimes used in the analyses below.

5.1.2 Latent Trajectory Dynamics

The purpose of analysing the latent trajectory dynamics is to determine whether the gap between logical and semantic composition appears only at the decoded endpoint or is already present during denoising. To do this, we compare semantic composition (e.g generating complex through a single prompt) and logical composition under a shared-noise protocol. The analysis will be organised by taxonomy group so that H1 is evaluated with respect to structured compositional regimes rather than a pooled average over heterogeneous pairs.

For each taxonomy group, we select six concept pairs and run twenty-four shared-noise seeds per pair. Each pair-seed record uses the same initial latent state for solo c_1, solo c_2, the monolithic prompt, and PoE. Seeds are treated as repeated measurements within a pair, so pair-level summaries remain the unit of group comparison.

This analysis asks three questions: whether the two trajectories remain close or diverge, when that divergence appears during denoising, and whether the expected Group 1–4 ordering is visible in the terminal gap. If the largest and earliest divergences concentrate in the structurally hardest groups, then H1 is supported at the level of latent dynamics rather than only decoded outputs.

This sets up the reachability analysis below.

5.1.3 Reachability Analysis

In this phase, we ask whether the terminal state of logical composition can be reproduced by prompt-conditioned generation. For each pair-seed record, the final composed image is inverted into a continuous prompt text embedding using SD-IPC [Ding *et al.* 2023b], and the model is rerun from the same initial noise under that recovered conditioning.

If the regenerated trajectory returns to logical composition's region, the gap may reflect a limitation of prompt choice within the semantic manifold. If it instead drifts back toward nearby semantic or dominant-concept modes, that supports the stronger claim that logical composition reaches targets that standard prompting does not readily recover. We compare this behaviour across the same Group 1–4 taxonomy, using trajectory closeness, terminal distance to the original PoE run, and output-level similarity as the main indicators.

5.1.4 Evaluation

To evaluate Hypothesis 1 at three complementary levels, we first measure the pair-level latent gap between semantic composition and PoE throughout denoising. We then compare their pooled terminal latent distributions within each taxonomy group. Finally, we test whether PoE endpoints can be recovered through prompt-conditioned regeneration using SD-IPC.

Latent-gap metric. We measure the composability gap as the latent distance between semantic composition and logical composition (PoE) at each denoising step:

$$d_t^{(p,s)} = \frac{1}{N} \left\| z_t^{\mathrm{sem},(p,s)} - z_t^{\mathrm{poe},(p,s)} \right\|_2^2, \qquad d_T^{(p,s)} = d_t^{(p,s)}\big|_{t=T}.$$

where p indexes the concept pair, s indexes the shared-noise seed, and N is the number of latent elements. Each $d_t^{(p,s)}$ is a seed-level latent gap: for one concept pair p and one shared-noise seed s, it measures the

mean squared difference between the semantic-composition latent and the PoE latent at denoising step t. The terminal quantity $d_T^{(p,s)}$ is the same measurement taken at the final denoising step, so it summarises the endpoint mismatch for that pair-seed run. Smaller values mean semantic and logical composition stay close in latent space; larger values indicate a stronger composability gap.

For each pair, we average over the 24 seeds:

$$\bar{d}_t^{(p)} = \frac{1}{24}\sum_{s=1}^{24} d_t^{(p,s)}, \qquad \bar{d}_T^{(p)} = \frac{1}{24}\sum_{s=1}^{24} d_T^{(p,s)}.$$

These averages give a more stable pair-level estimate of how that concept pair behaves overall, rather than how it behaved under any single random initialisation.

Temporal divergence. To determine when the gap emerges, we divide the 50-step denoising trajectory into early, mid, and late phases and measure how much the latent gap grows in each phase:

$$\Delta_{\text{early}}^{(p,s)} = d_{17}^{(p,s)} - d_0^{(p,s)}, \qquad \Delta_{\text{mid}}^{(p,s)} = d_{34}^{(p,s)} - d_{17}^{(p,s)}, \qquad \Delta_{\text{late}}^{(p,s)} = d_{50}^{(p,s)} - d_{34}^{(p,s)}.$$

These quantities are phase-wise growth terms. For example, $\Delta_{\text{early}}^{(p,s)} = d_{17}^{(p,s)} - d_0^{(p,s)}$ measures how much the gap between semantic composition and PoE increases between the start of denoising and step 17 for one concept pair p under one shared-noise seed s. Similarly, $\Delta_{\text{mid}}^{(p,s)}$ measures how much the gap grows between steps 17 and 34, and $\Delta_{\text{late}}^{(p,s)}$ measures how much additional gap is accumulated between steps 34 and 50. In other words, these terms do not measure the total gap itself, but how much new divergence is added during each phase of denoising.

This lets us ask not only *how large* the composability gap is, but also *when* it appears. If $\Delta_{\text{early}}^{(p,s)}$ is already large, then semantic and logical composition separate early, suggesting that the incompatibility is injected during the high-noise structural phase. If most of the growth appears in $\Delta_{\text{late}}^{(p,s)}$, then the two mechanisms remain relatively close until later refinement stages. Under H1, the harder groups should show larger early and total growth than the easier groups.

Group-level distribution gap. The latent-gap metric compares semantic and PoE one pair at a time. To compare them at the level of whole taxonomy groups, we pool all pair-seed records within each group g and compare the resulting semantic and PoE distributions:

$$P_g^{\text{sem}} \text{ for semantic composition in group } g, \qquad P_g^{\text{poe}} \text{ for PoE in group } g.$$

Here, P_g^{sem} denotes the pooled semantic-composition distribution within group g, and P_g^{poe} denotes the corresponding pooled PoE distribution. Intuitively, instead of asking whether semantic and PoE differ for one pair at a time, we now ask whether they occupy different overall distributional regimes across the group as a whole.

Because KL divergence is asymmetric, we use Jeffreys divergence as a symmetric summary:

$$J_g = D_{\text{KL}}\big(P_g^{\text{sem}} \,\|\, P_g^{\text{poe}}\big) + D_{\text{KL}}\big(P_g^{\text{poe}} \,\|\, P_g^{\text{sem}}\big)\,.$$

Small J_g means the pooled semantic and PoE group distributions are similar. Large J_g means they differ more strongly, indicating a larger group-level composability gap. The expected ordering is

$$J_{G1} \approx J_{G2} < J_{G3} < J_{G4}.$$

Reachability. To test whether the PoE target is reachable through language-conditioned regeneration, we recover an SD-IPC prompt embedding $p^\star$, rerun the model from the same initial noise, and measure the residual distance to the original PoE trajectory:

$$d_t^{\star,(p,s)} = \frac{1}{N}\left\| z_t^{\star,(p,s)} - z_t^{\text{poe},(p,s)} \right\|_2^2, \qquad d_T^{\star,(p,s)} = d_t^{\star,(p,s)}\big|_{t=T}.$$

Each $d_t^{\star,(p,s)}$ is a seed-level reachability residual: for one concept pair p and one shared-noise seed s, it measures the mean squared difference between the rerun trajectory induced by the recovered prompt

embedding $p^\star$ and the original PoE trajectory at denoising step t. The terminal quantity $d_T^{\star,(p,s)}$ is the same residual measured at the final denoising step, so it summarises how far the rerun endpoint remains from the original PoE target for that pair-seed run.

Intuitively, this metric asks whether a language-conditioned rerun can return to the same latent region reached by logical composition. If the recovered prompt embedding $p^\star$ captures the PoE target well, then the rerun trajectory should stay close to the original PoE trajectory, giving small residuals. If the rerun drifts away from the PoE path, then the residual grows, indicating that the logical target is not fully recoverable through standard prompting alone.

Smaller residuals therefore mean the prompt-conditioned rerun remains close to the PoE target, while larger residuals mean the logical target is not fully reachable through language-conditioned regeneration alone.

H1 is supported if the gap is measurable, emerges earlier and more strongly in harder groups, Jeffrey divergence follows the taxonomy ordering, and PoE remains less recoverable by prompt-conditioned reruns in Groups 3 and 4. H1 is weakened if only some of these metrics show the expected ordering. H1 is not supported if semantic and PoE remain close across groups and prompt-conditioned recovery succeeds broadly.

5.1.5 Summary

Phase 1 characterises the composability gap as a structured empirical phenomenon. The taxonomy identifies the concept regimes in which the gap should be minimal or severe, the trajectory analysis tests whether semantic and logical composition separate during denoising, and the reachability analysis tests whether the endpoint reached by logical composition remains recoverable through prompt-conditioned generation. Taken together, these analyses support H1 when the gap worsens predictably from Groups 1–2 to Group 4, with the largest and earliest divergence appearing in the hardest regimes.

What Phase 1 does not resolve is the cause of this pattern. Because Stable Diffusion [Rombach *et al.* 2022] fixes both the internal representations and the score compositional method, these results cannot yet determine whether the gap is fundamentally a representation problem or a compositional method. Phase 2 addresses that question directly by testing the representation hypothesis in a controlled setting, asking whether our novel architecture that progressively disentangles latents reduces the same compositional failures observed here.

5.2 Phase 2: Explaining the Composability Gap

Phase 1 characterised the composability gap across concept regimes. Phase 2 asks whether that gap is partly a representation problem. The key claim is that if composition is limited by the quality of the learned representation, then more factorised latent spaces should yield more reliable composition under the same downstream conditional diffusion setup.

This distinction is difficult to test directly in a large pretrained model such as Stable Diffusion [Rombach *et al.* 2022], where the learned representation and the inference-time compositional method are tightly coupled. Phase 2 therefore moves to a controlled synthetic setting in which the representation can be varied while the downstream composition pipeline is kept fixed. This leads to a central claim:

> **If representation quality is a limiting factor for composition, then more factorised latent spaces should yield more reliable compositional generation under the same downstream conditional diffusion architecture.**

H2 is supported if better factorised latent spaces improve held-out composition under this shared evaluator.

5.2.1 Dataset Selection

We use dSprites [Matthey *et al.* 2017] and a correlated dSprites variant [Roth *et al.* 2023] because the underlying factors are known and held-out recombination can be measured directly. Standard dSprites serves as the positive control: orientation and scale are sampled independently, and one orientation–scale

combination is removed from training. If a model fails to recover that held-out combination in this setting, the limitation is already present before support bias is introduced.

Correlated dSprites: Correlated dSprites keeps the visual domain unchanged but biases orientation–scale co-occurrence. Some combinations become common while others are rare or missing, creating a controlled setting in which the model can rely on shortcut correlations rather than learning the factors separately. This makes it well suited for testing whether better latent factorisation restores held-out composition.

5.2.2 Experiment Design

The experiment compares four regimes under the same downstream evaluation. Regimes A and B establish the baseline effect of changing the training support, Regime C tests whether disentanglement or weak supervision recover performance in the correlated setting, and Regime D tests whether explicit latent partitioning improves further.

Regime A: Control

Regime A trains a baseline VAE on standard dSprites. It asks whether held-out recombination is feasible when the training support is factorised. Success here establishes that the task itself is valid before correlation is introduced.

Regime B: Correlated dSprites Setting

Regime B keeps the same baseline VAE but trains it on correlated dSprites. This isolates the effect of biased support: if performance drops relative to Regime A, then the failure can be attributed to support-induced entanglement rather than a change in model family or evaluation rule.

Regime C: Disentangling Correlated dSprites

Regime C keeps the correlated support of Regime B but strengthens representation learning in two ways. C1 uses standard disentanglement models such as β-VAE and FactorVAE. C2 adds weak supervision on orientation and scale through training-time auxiliary heads while leaving the latent space shared. This regime tests whether stronger independence pressure or limited supervision alone can recover held-out composition under correlation.

Regime D: Partitioned Representation

Regime D keeps the same correlated support and matched supervision as C2, but replaces the shared latent with a partitioned representation. The latent code is organised as $z = (z_c, z_{d1}, z_{d2})$, where z_c stores shared context, z_{d1} orientation, and z_{d2} scale. If Regime D improves beyond C2, the gain is most plausibly due to explicit factor-addressable structure rather than supervision alone.

5.2.3 Evaluation

Phase 2 evaluates H2 in two stages. First, it asks whether each regime learns a latent representation that is more clearly aligned with the known dSprites factors. Second, it asks whether any representational improvement translates into better recovery of withheld orientation–scale combinations under the same downstream conditional latent diffusion setup. The first stage therefore evaluates representation quality; the second evaluates whether that representation is actually useful for recombination.

DCI. We use DCI [Eastwood and Williams 2018; Locatello *et al.* 2019] as the first-stage representation metric because it measures how strongly the learned latent code aligns with the ground-truth generative factors. This is the central representational question in Phase 2. All regimes are trained on the same visual domain and are evaluated on the same downstream recombination task, so DCI provides a common probe-based summary of how factor-aligned each learned code is.

DCI is useful here because its three components answer three distinct questions. *Disentanglement* asks whether an individual latent dimension is used mainly for one factor rather than several. *Completeness* asks whether a given factor is concentrated in a small part of the code rather than spread broadly across

many dimensions. *Informativeness* asks whether the code preserves enough information to recover the true factors at all. Taken together, these scores distinguish a code that is merely predictive from one that is also well organised.

The regime-level interpretation is direct. In Regime A, DCI acts as a positive-control check under factorised support. In Regime B, it tests whether correlated support weakens factor alignment. In Regime C, it tests whether stronger disentanglement objectives or weak supervision recover that alignment. In Regime D, it tests whether explicit latent partitioning yields a more factor-addressable code than a single shared latent space.

For DCI, each evaluation image $x^{(i)}$ is encoded into a deterministic representation

$$h^{(i)} = (h_1^{(i)}, \dots, h_K^{(i)}).$$

We use a deterministic code rather than a sampled latent so that the evaluation is not affected by reparameterisation noise. Concretely, for Regimes A–C we set $h^{(i)} = \mu_\phi(x^{(i)})$, the encoder posterior mean. For Regime D, whose latent is partitioned as $z = (z_c, z_{d1}, z_{d2})$, we use the concatenated posterior means

$$h^{(i)} = \big[\mu_c(x^{(i)}), \mu_{d1}(x^{(i)}), \mu_{d2}(x^{(i)})\big].$$

This keeps the DCI input deterministic and comparable across regimes while preserving Regime D's structural allocation.

Each example also has known ground-truth factor labels

$$c^{(i)} = (c_1^{(i)}, \dots, c_J^{(i)}).$$

We compute DCI using all ground-truth dSprites factors available in the dataset, namely shape, scale, orientation, and position, while focusing the interpretation on orientation and scale because these are the factors used in the withheld recombination task. This avoids overstating factor separation by evaluating only the target pair.

Following Eastwood and Williams [2018], we fit one supervised predictor per factor using the latent code h as input. We use gradient-boosted tree predictors throughout because they can model nonlinear latent–factor relations while also yielding a feature-importance score for each latent dimension. The same probe family, training split, and hyperparameter-selection procedure are used for every regime so that DCI differences reflect changes in the learned representation rather than changes in the evaluator.

Let R_{kj} denote the importance assigned to latent dimension h_k when predicting factor c_j. Collecting these importances across all factors yields the importance matrix

$$R = \begin{bmatrix} R_{11} & R_{12} & \cdots & R_{1J} \\ R_{21} & R_{22} & \cdots & R_{2J} \\ \vdots & \vdots & \ddots & \vdots \\ R_{K1} & R_{K2} & \cdots & R_{KJ} \end{bmatrix}.$$

Each column shows which latent dimensions matter for one factor, while each row shows how one latent dimension is used across factors. After fitting the factor predictors, we extract their feature importances, place them into R, and then summarise the concentration structure of the rows and columns.

For intuition, consider only the two target factors, orientation and scale. A row such as $(0.90, 0.10)$ indicates that one latent dimension contributes strongly to orientation prediction and only weakly to scale prediction. A different row such as $(0.10, 0.85)$ indicates the opposite. This is the kind of pattern expected from a well-factorised representation. By contrast, a row such as $(0.50, 0.50)$ suggests that the same latent dimension is being used for both factors, which is less desirable if the goal is clean factor separation.

DCI disentanglement asks whether the predictive role of each latent dimension is concentrated on one factor or spread across several. For each latent dimension h_k, we therefore normalise its importance scores across factors:

$$D = \sum_{k=1}^{K} \rho_k \big(1 - H(P_k)\big), \qquad P_{kj} = \frac{R_{kj}}{\sum_{j'} R_{kj'}}, \qquad \rho_k = \frac{\sum_j R_{kj}}{\sum_{k',j} R_{k'j}}.$$

Here P_k is the normalised k-th row of R, so P_{kj} gives the fraction of latent dimension h_k's total predictive importance that is assigned to factor c_j. The quantity $H(P_k)$ is the entropy of that row, normalised to lie in $[0, 1]$. Low entropy means that the row is concentrated on one factor, so that latent contributes strongly to disentanglement. High entropy means that the same latent is used more evenly across several factors, so its disentanglement contribution is lower. The weight ρ_k ensures that latents which contribute very little overall do not dominate the score.

DCI completeness asks the complementary question: for a fixed factor, is its predictive signal concentrated in a small number of latent dimensions or spread across many? For each factor c_j, we normalise the corresponding column of the importance matrix:

$$C = \frac{1}{J}\sum_{j=1}^{J}\big(1 - H(Q_j)\big), \qquad Q_{jk} = \frac{R_{kj}}{\sum_{k'} R_{k'j}}.$$

Here Q_j is the normalised j-th column of R, so Q_{jk} measures how much of factor c_j's predictive signal comes from latent dimension h_k. If the orientation column is dominated by one latent dimension and the remaining latents contribute only weakly, completeness for orientation is high. If orientation is instead spread broadly across many latents, completeness is lower. In plain terms, completeness is high when a factor is concentrated in a small part of the code rather than distributed diffusely across it.

DCI informativeness measures how well the latent code predicts the true factors overall. In our setting, the factors are discrete, so we report informativeness using held-out classification accuracy for each factor predictor, together with the mean accuracy across factors. This matters because a code can appear tidy while still failing to preserve enough information to recover the true attributes. Informativeness therefore checks whether the representation is actually usable, not only whether it appears structurally clean.

Taken together, DCI asks whether each latent dimension mainly captures one factor, whether each factor is concentrated in a small number of latent dimensions, and whether the code preserves enough information to recover the true factors at all. If correlated support causes the baseline VAE to mix orientation and scale across the code, disentanglement and completeness should decline relative to Regime A. If later regimes restore clearer factor alignment, those scores should improve. DCI therefore serves as the representation-level half of the Phase 2 evaluation, with the downstream diffusion-based recombination analysis testing whether that representational gain translates into better compositional generation.

Held-out recombination accuracy. Held-out recombination accuracy evaluates whether generated images recover withheld orientation–scale combinations under the same downstream evaluation setup used for all Phase 2 regimes. Here, a *withheld combination* means an orientation–scale pair that is deliberately removed from the training set, so that successful generation requires recombining factors in a way not observed during training.

To score the generated samples, we use two *external classifiers*: an image-level orientation classifier and an image-level scale classifier trained separately from the representation-learning model and used only for evaluation. For each generated test image, the orientation classifier predicts $\hat{y}^{(i)}_{\text{orient}}$ and the scale classifier predicts $\hat{y}^{(i)}_{\text{scale}}$. These predictions are then compared against the target withheld labels $y^{(i)}_{\text{orient}}$ and $y^{(i)}_{\text{scale}}$.

We first report two per-factor accuracies:

$$\text{Acc}_{\text{orient}} = \frac{1}{N}\sum_{i=1}^{N} r^{(i)}_{\text{orient}}, \qquad \text{Acc}_{\text{scale}} = \frac{1}{N}\sum_{i=1}^{N} r^{(i)}_{\text{scale}},$$

where

$$r^{(i)}_{\text{orient}} = \begin{cases} 1, & \text{if } \hat{y}^{(i)}_{\text{orient}} = y^{(i)}_{\text{orient}}, \\ 0, & \text{otherwise,} \end{cases} \qquad r^{(i)}_{\text{scale}} = \begin{cases} 1, & \text{if } \hat{y}^{(i)}_{\text{scale}} = y^{(i)}_{\text{scale}}, \\ 0, & \text{otherwise.} \end{cases}$$

These scores indicate whether each target factor is recovered individually across the generated test set. For example, if the withheld combination is a particular orientation paired with a particular scale, $\text{Acc}_{\text{orient}}$ measures how often the generated images receive the correct orientation label, while $\text{Acc}_{\text{scale}}$ measures how often they receive the correct scale label.

The average of the per-factor accuracies does not measure held-out recombination, because it does not require the correct orientation and correct scale to occur in the same generated sample. We therefore

use joint recovery accuracy as the main downstream metric, and report per-factor accuracies only as diagnostic summaries:

$$\mathrm{Rec}_{\mathrm{joint}} = \frac{1}{N}\sum_{i=1}^{N} r_{\mathrm{joint}}^{(i)},$$

where

$$r_{\mathrm{joint}}^{(i)} = \begin{cases} 1, & \text{if } \hat{y}_{\mathrm{orient}}^{(i)} = y_{\mathrm{orient}}^{(i)} \text{ and } \hat{y}_{\mathrm{scale}}^{(i)} = y_{\mathrm{scale}}^{(i)}, \\ 0, & \text{otherwise.} \end{cases}$$

A simple example clarifies this distinction. Suppose a particular large-scale, left-rotated sprite combination is withheld during training. If a generated image is classified as left-rotated but small, then orientation is correct and scale is incorrect. That sample contributes positively to $\mathrm{Acc}_{\mathrm{orient}}$, negatively to $\mathrm{Acc}_{\mathrm{scale}}$, and contributes $r^{(i)} = 0$ to $\mathrm{Rec}_{\mathrm{joint}}$. Only a generated image for which both classifier predictions match the intended withheld labels counts as a successful recombination.

High per-factor accuracy shows that the generated images preserve the target factors individually. High joint recovery accuracy shows that those factors are recovered together in the same image, which is the central requirement for held-out recombination. Conversely, low joint recovery accuracy, especially when the per-factor scores are higher, indicates partial recovery without successful joint recombination.

This metric has a specific scope. It evaluates whether a regime can generate unseen factor combinations under the shared downstream setup. It does not by itself measure perceptual realism, reconstruction fidelity, or latent disentanglement, and it depends on the reliability of the external classifiers used for scoring. For that reason, held-out recombination accuracy should be interpreted as a targeted downstream test of out-of-distribution factor recovery in the generated samples.

5.2.4 Summary

Phase 2 addresses the gap identified in Phase 1 by testing H2 in a controlled synthetic setting. Regime A provides the control, Regime B creates the failure setting, Regime C tests whether disentanglement or weak supervision recover performance, and Regime D tests whether explicit latent partitioning improves further. If performance falls from Regime A to Regime B and then improves most clearly in Regime D, Phase 2 supports the claim that representation quality is a meaningful limiting factor for composition.

Phase 3 then asks whether the same idea remains useful for chest X-rays, where conditional independence is approximate and success must be judged by radiological plausibility.

5.3 Phase 3: Compositional Generation of Co-morbid Chest X-rays

This phase is the clinical extension of the representation hypothesis developed in Phases 1 and 2. Phase 1 examines when logical composition fails, and Phase 2 tests whether better-separated representations improve held-out composition in a controlled environment. Building on this, Phase 3 asks whether the same basic idea can be used to generate clinically plausible co-morbid chest X-rays.

We study posteroanterior (PA) chest X-rays containing multiple disease conditions and treat co-morbid synthesis as an out-of-distribution composition problem. The central question is whether the selected diseases are anatomically distinct enough that their effects can be represented as approximately factorized conditionals over shared anatomy. Under this assumption, composition may produce clinically plausible co-morbid images even when joint cases are scarce.

Evaluation focuses on pathology presence, spatial plausibility, radiological fidelity, and held-out co-morbid composition. This phase matters because real co-morbid cases are rare and expensive to annotate. If successful, it would show that compositional generation can help synthesise clinically plausible examples of underrepresented disease combinations.

5.3.1 Dataset Selection

This phase requires a clinical dataset with enough PA chest X-rays, reliable pathology annotations, and meaningful class imbalance to make rare co-morbidity a realistic problem. VinBigData [Nguyen *et al.*

2020] Chest X-ray is well suited to this setting because it provides 15,000 PA radiographs with radiologist annotations across 14 findings in a strongly long-tailed distribution. Its pathology labels and bounding boxes make it suitable both for selecting a scarce pathology pair and for checking whether disease evidence remains anatomically localised.

As illustrated in Figure 5.1, the dataset is dominated by "No finding" cases, with the remaining pathologies spread across a pronounced long tail. This matters because disease-specific signal can easily be overwhelmed by normal anatomy if the data are used naively. For that reason, training will use balanced sampling across the selected pathologies so that rare findings are not suppressed during learning.

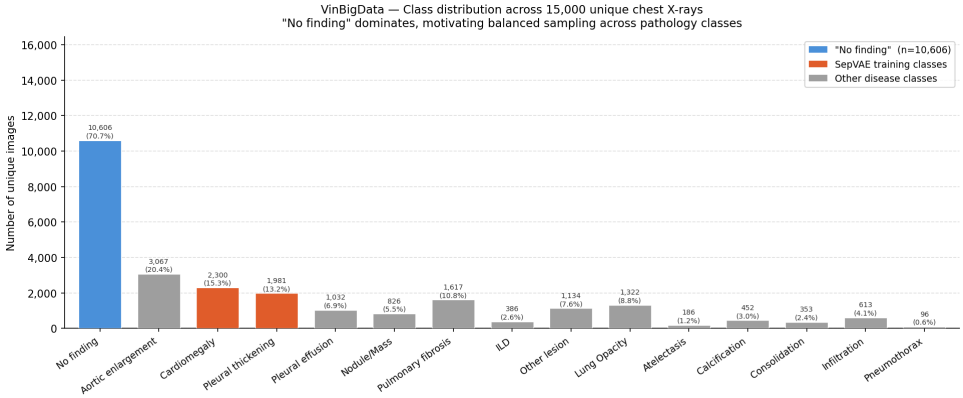


Figure 5.1: Class distribution across 15,000 chest X-rays in VinBigData. The dominance of "No finding" highlights the long-tailed nature of the dataset and motivates balanced sampling during training.

We focus on cardiomegaly and pleural thickening, where cardiomegaly primarily affects the central heart region and pleural thickening appears along the pleural boundary and lung periphery. This pair is clinically plausible because both findings can co-occur in real patients, yet it remains compositionally nontrivial because their joint presentation is relatively rare and both disease effects must be represented within the same underlying chest anatomy. Successful synthesis therefore requires the model to preserve shared anatomical structure while expressing each pathology distinctly, without suppression, leakage, or anatomically implausible blending.

As shown in Figure 5.2, each pathology occurs often enough to support separate learning, but their joint presentation is considerably less frequent: cardiomegaly appears in 2,300 images, pleural thickening in 1,981, and the two are annotated together in only 858. This creates a realistic low-resource co-morbidity setting in which a standard monolithic model sees many examples of each disease in isolation but far fewer examples of how they appear together.

Figure 5.2 therefore defines the low-resource regime in which Phase 3 operates. The model has access to many single-disease examples but comparatively few examples of their joint presentation, so the experiment asks whether the abundant single-disease cases can be used as separate priors and composed to synthesise plausible co-morbid radiographs.

5.3.2 Experiment Design

Preprocessing

Phase 3 begins by matching each raw DICOM radiograph to its pathology labels and bounding-box annotations. VinBigData contains radiographs acquired across multiple clinical sites, so the images are first converted to a common representation before resizing and normalisation. This reduces nuisance variation from acquisition and makes disease-related structure more comparable across patients and clinical sites.

Bounding-box annotations are then retained as weak spatial guidance. They serve two purposes in the proposal. First, they provide an exploratory view of where each selected pathology is typically expressed in the image. As shown in Figure 5.3, cardiomegaly is concentrated around the enlarged cardiac silhouette in the central chest, while pleural thickening appears along the lung periphery and lower pleural margins. Second, they provide weak spatial supervision during representation learning, helping align disease-related

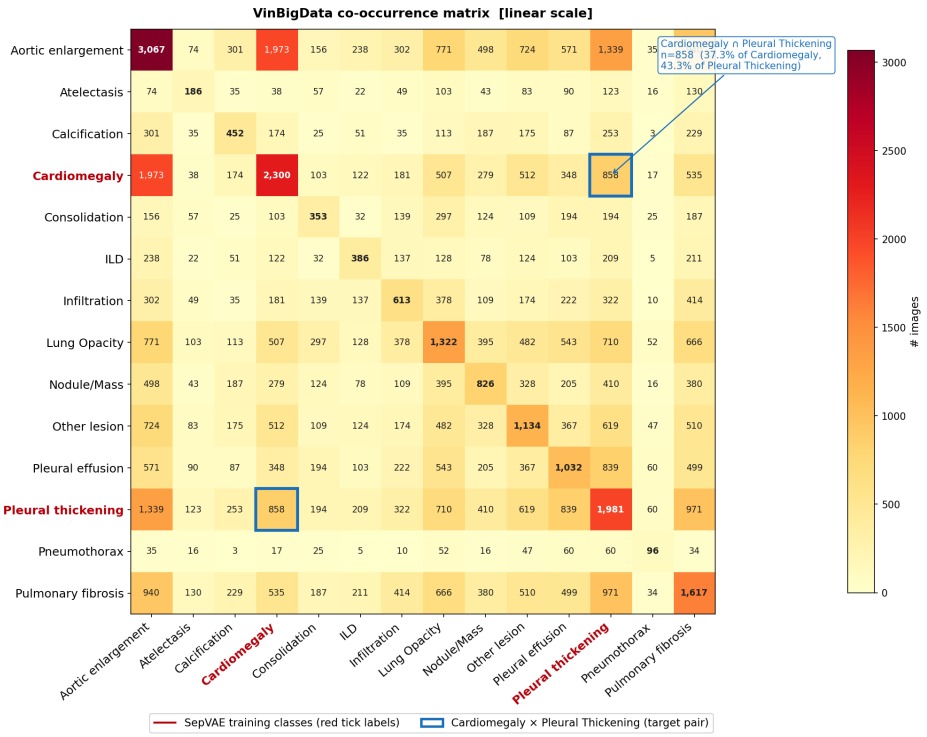


Figure 5.2: VinBigData co-occurrence and scarcity heatmap. The bright diagonal reveals individual disease prevalence, while the darker off-diagonal cells highlight the low-resource co-morbidity problem.

variation with the expected anatomical regions rather than letting it spread diffusely across the chest. This is intended to give the downstream diffusion model a more spatially structured latent representation.

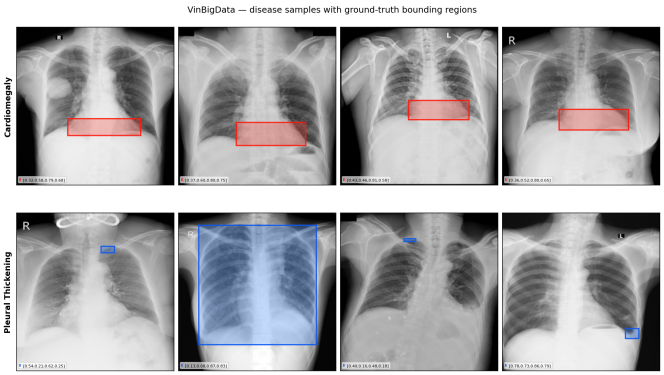


Figure 5.3: Ground-truth examples illustrating the spatial distinctness of Cardiomegaly and Pleural Thickening in VinBigData. Cardiomegaly is concentrated around the cardiac silhouette, whereas pleural thickening is expressed along the lung periphery and pleural margin.

Figure 5.3 does not by itself establish compositional separability, but it shows why this pathology pair is a plausible first test case. The dominant visual evidence for the two diseases tends to occupy different anatomical substrates, which provides a basis for testing whether approximate spatial separation can be turned into a more factorised latent representation.

Conditional Assumptions

In Phase 3, composition is not performed against a fully unconditional latent distribution. Instead, it is performed relative to a shared anatomical scaffold z_c, which is intended to capture patient-specific and acquisition-related structure common to both disease conditions. The diffusion variable u then represents the residual generative state over which disease-specific variation is modeled. Conditioning on z_c therefore changes the reference distribution in Bayes composition from $p(u)$ to $p(u \mid z_c)$: rather than combining

cardiomegaly and pleural-thickening evidence over arbitrary chest X-rays, we combine them over a fixed shared anatomical context. Under the assumption

$$c_1 \perp c_2 \mid (u, z_c),$$

this gives

$$p(u \mid z_c, c_1, c_2) \;\propto\; \frac{p(u \mid z_c, c_1)\, p(u \mid z_c, c_2)}{p(u \mid z_c)}. \tag{5.1}$$

In this setting, c_1 and c_2 correspond to cardiomegaly and pleural thickening respectively. This is not a claim that cardiomegaly and pleural thickening occur independently in clinical data. Rather, it is the modelling assumption that once the shared context z_c and generative state u are fixed, the two disease labels can be treated as approximately independent for the purpose of composition. The three assumptions below motivate when this conditional independence is plausible: the selected diseases are approximately separable in practice, shared anatomy can be factorised from disease-specific effects, and under that factorisation the remaining disease conditionals can be treated as approximately independent given (u, z_c).

1. **The selected diseases are approximately separable in practice.** Cardiomegaly primarily affects the heart region, whereas pleural thickening appears along the pleural boundary and lung periphery. This spatial distinction does not by itself guarantee compositional success, but it provides practical evidence that the two disease effects may be separable in a suitable representation.

2. **Shared anatomy and disease-specific effects can be factorised.** Given this approximate anatomical separability, we assume the representation can be decomposed into shared anatomy and disease-specific components. In particular, common chest anatomy and acquisition variation can be separated from disease-related variation, and the two disease effects can be represented with limited leakage into one another.

3. **These separations induce approximately factorised conditionals over shared anatomy.** If the first two assumptions hold, then the single-disease conditionals can be treated as approximately factorised over a shared anatomical background. Under this assumption, Bayes-style composition becomes a plausible approximation for generating clinically plausible co-morbid images when joint cases are scarce.

Taken together, these assumptions define the domain in which Phase 3 is expected to succeed. They do not claim that co-morbid composition is guaranteed in clinical data; rather, they identify a setting in which Bayes-style composition is theoretically plausible enough to test. The architecture and training objective that follow are designed to encourage a representation consistent with these assumptions, so that held-out co-morbid synthesis becomes an empirical test of H3 rather than an unconstrained application of score addition.

Figure 5.4 shows that, although the two diseases can co-occur clinically, their dominant visual evidence is concentrated in different anatomical regions. Cardiomegaly is centred on the heart region, whereas pleural thickening is concentrated along the peripheral lower pleural margins.

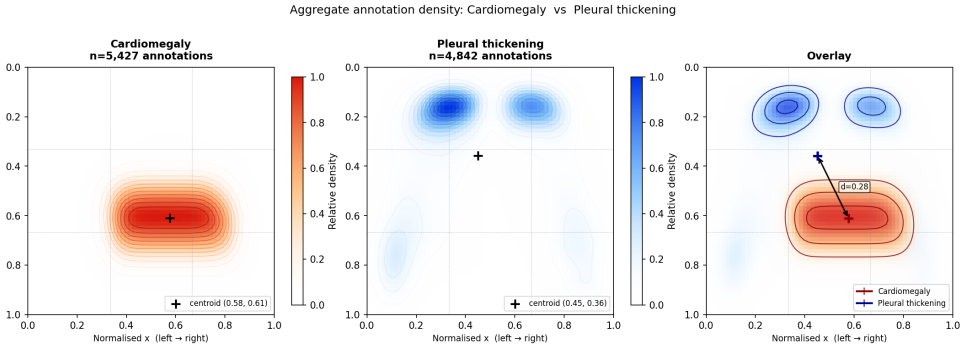


Figure 5.4: Aggregate annotation density for Cardiomegaly (n = 2,300) and Pleural Thickening (n = 1,981), showing distinct hotspots in the central heart region and along the peripheral lower pleural margins respectively.

Representation Learning

These assumptions motivate an explicit architectural intervention rather than relying on a monolithic latent to disentangle anatomy and pathology implicitly. Phase 3 therefore introduces a partitioned clinical VAE inspired by SepVAE's common versus salient partitioning, but adapted to a single-dataset co-morbidity setting. Instead of separating background and target images across two datasets, the model allocates separate latent roles for shared structure and for each disease-specific effect within the same chest X-ray.

This matters because a monolithic latent can still exploit co-occurrence shortcuts and encode both diseases jointly rather than compositionally. Even with supervision, cardiomegaly and pleural thickening could still leak into the same latent dimensions if that improves reconstruction. To address this, Phase 3 introduces a partitioned code,

$$z = (z_c,\, z_{d1},\, z_{d2}), \tag{5.2}$$

where z_c captures shared anatomy, acquisition context, and other non-pathological variation, while z_{d1} and z_{d2} capture the two disease-specific factors. In the present setting, z_{d1} is aligned with cardiomegaly-specific evidence and z_{d2} with pleural-thickening-specific evidence. The encoder therefore produces three posterior heads, one for each latent part, whose samples are concatenated and decoded into the reconstructed radiograph. In the compositional formulation below, z_c provides the shared scaffold, while disease-specific effects are represented so they can be composed conditionally over that scaffold.

This latent partition is complemented by training-only auxiliary predictor heads applied to the encoder means of the disease latents. One classifier is trained to recover cardiomegaly from μ_{d1} and the other to recover pleural thickening from μ_{d2}, aligning each latent branch with its intended pathology. To discourage cross-factor leakage, the same predictors are also applied to the opposite latent branch through gradient reversal layers (GRLs), so that the classifiers detect residual leakage while the encoder is trained to remove it. These auxiliary heads are discarded at test time, leaving a generative model whose composition interface is the partitioned latent representation itself.

At a high level, the objective function is composed of the following:

- **Reconstruction and regularisation:** these preserve faithful reconstruction while keeping the shared and disease-specific latent blocks well behaved.
- **Correct-head supervision:** auxiliary predictors applied to μ_{d1} and μ_{d2} encourage each disease-specific latent to encode its intended pathology, making the partition clinically interpretable rather than merely structural.
- **Wrong-head suppression with GRLs:** auxiliary predictors applied to the opposite latent branch through gradient reversal layers discourage cardiomegaly evidence from leaking into z_{d2} and pleural-thickening evidence from leaking into z_{d1}.
- **Dependence reduction:** orthogonality or mutual-information penalties reduce residual statistical and geometric overlap between z_{d1} and z_{d2} beyond what the classifier-based terms remove.
- **Region-weighted reconstruction:** bounding-box-guided reconstruction promotes locality by forcing disease-related changes to be explained primarily within the annotated pathology regions.

Together, these terms preserve a usable generative model while pushing the learned representation toward factor-addressable disease latents. The intended outcome is that shared anatomy remains in z_c, each disease-specific signal is captured in the appropriate latent part, and the two factors can later be combined with less interference at inference time.

Latent Diffusion Model

The partitioned VAE defines the representation, but the actual generative composition is performed by a single conditional latent diffusion model trained on that latent space. We write u_0 for the clean generative variable modeled by the diffusion process and u_t for its noisy version at time t, with generation initialized from $u_T \sim \mathcal{N}(0, I)$. In this setup, diffusion operates over the generative variable u, while z_c provides the shared anatomical context. Composition is performed over u conditioned on z_c, and the null condition $\varnothing$ corresponds to omitting the disease label while keeping that shared context fixed.

We do not train separate unconditional and conditional generators, because classifier-free guidance assumes a single conditional diffusion model that can be evaluated under different conditions. During

training, each u_0 is paired with its observed disease label $y \in \{c_1, c_2\}$, and that label is replaced by the null condition $\varnothing$ with probability p_{uncond}. This gives one diffusion model that learns the conditional branches $\varepsilon_\theta(u_t, t \mid z_c, c_1)$ and $\varepsilon_\theta(u_t, t \mid z_c, c_2)$ together with the branch $\varepsilon_\theta(u_t, t \mid z_c, \varnothing)$. Here, $\varnothing$ means no disease label given z_c, not a literal normal or no-finding class. In the low-resource setting of Phase 3, joint cardiomegaly and pleural-thickening cases are absent or heavily reduced during generative training, but the model still learns the two single-disease conditionals and the null branch required for conditional composition.

Under the conditional independence assumption above, the corresponding score decomposition is

$$\nabla_u \log p(u \mid z_c, c_1, c_2) = \nabla_u \log p(u \mid z_c, c_1) + \nabla_u \log p(u \mid z_c, c_2) - \nabla_u \log p(u \mid z_c). \tag{5.3}$$

In practice, Phase 3 implements the classifier-free guidance analogue of this conditional score decomposition:

$$\begin{aligned} \tilde{\varepsilon}_{\text{AND}}(u_t, t; z_c) = \varepsilon_\theta(u_t, t \mid z_c, \varnothing) \\ + w_1 \big[\varepsilon_\theta(u_t, t \mid z_c, c_1) - \varepsilon_\theta(u_t, t \mid z_c, \varnothing)\big] \\ + w_2 \big[\varepsilon_\theta(u_t, t \mid z_c, c_2) - \varepsilon_\theta(u_t, t \mid z_c, \varnothing)\big]. \end{aligned} \tag{5.4}$$

This ε-based form is the practical CFG-style approximation to the conditional Bayes score decomposition, not an exact probabilistic identity. The resulting denoised latent is then decoded through the VAE decoder to produce a candidate co-morbid radiograph, which is passed to the evaluation procedure below. In this way, Phase 3 combines a partitioned representation with a single conditional latent diffusion model that implements Bayes-style composition over a shared scaffold.

5.3.3 Evaluation

Phase 3 evaluates whether a factorised clinical representation supports *held-out co-morbid composition* in a realistic low-resource setting. Here, *co-morbidity* means the simultaneous presence of the two target pathologies in the same chest radiograph, and a *held-out co-morbid presentation* means a joint cardiomegaly–pleural-thickening case that is absent or strongly reduced during generative training, even though single-disease cases remain available. The goal is therefore not only to generate visually plausible images, but to test whether the model can recover a clinically meaningful joint presentation that was not available, or was severely underrepresented, during training.

In this phase, *composition* refers to combining the two disease-specific latent factors over a shared anatomical scaffold z_c and decoding the result into a candidate co-morbid radiograph. This is the clinical analogue of held-out recombination in Phase 2, but the target is now a realistic joint disease presentation rather than a synthetic attribute pair. The evaluation therefore tests two related questions: first, whether the model can generalise to an unseen or severely underrepresented co-morbid combination; and second, whether the resulting sample remains radiologically plausible rather than collapsing into an artificial overlay or a spatially incoherent hybrid.

To evaluate this, each composed image x is assessed along three dimensions: pathology presence, co-morbidity fidelity, and spatial plausibility. These three criteria jointly define success in the clinical setting. Pathology presence checks whether recoverable evidence for both target diseases is present in the generated image. Co-morbidity fidelity checks whether the image resembles a genuine joint presentation rather than merely containing two isolated disease cues. Spatial plausibility checks whether the disease evidence appears in the anatomically expected regions motivated by the pathology annotations.

Pathology presence. We first test whether the composed image contains recoverable evidence for both target diseases. To do this, we use two external pathology classifiers, trained on data splits disjoint from generative-model training: a cardiomegaly classifier $f_{\text{card}}(x)$ and a pleural-thickening classifier $f_{\text{pleural}}(x)$, each of which returns a probability of pathology presence.

For a composed sample $x^{(i)}$, binary pathology recovery is defined as

$$p^{(i)}_{\text{card}} = \begin{cases} 1, & \text{if } f_{\text{card}}(x^{(i)}) \geq \tau_{\text{card}}, \\ 0, & \text{otherwise}, \end{cases} \qquad p^{(i)}_{\text{pleural}} = \begin{cases} 1, & \text{if } f_{\text{pleural}}(x^{(i)}) \geq \tau_{\text{pleural}}, \\ 0, & \text{otherwise}. \end{cases}$$

Here τ_{card} and τ_{pleural} are predefined decision thresholds that convert the pathology-classifier output probabilities into binary presence decisions, calibrated on a validation split and fixed before final testing.

Joint pathology presence is then

$$p_{\text{joint}}^{(i)} = \begin{cases} 1, & \text{if } p_{\text{card}}^{(i)} = 1 \text{ and } p_{\text{pleural}}^{(i)} = 1, \\ 0, & \text{otherwise.} \end{cases}$$

The overall pathology-presence rate is

$$\text{Pres} = \frac{1}{M} \sum_{i=1}^{M} p_{\text{joint}}^{(i)}.$$

This is the clinical analogue of factor recovery in Phase 2: it asks whether both target pathologies remain recoverable in the generated samples, not merely whether the image appears plausible overall.

Co-morbidity fidelity. Pathology presence alone is not sufficient, because an image could score positive for both pathologies yet still fail to resemble a genuine co-morbid radiograph. To test whether the composed image looks like a plausible joint presentation rather than an artificial overlay, we use a third external classifier,

$$f_{\text{co}}(x^{(i)}) \in [0, 1],$$

trained on real co-morbid chest X-rays. Its output is interpreted as a co-morbidity fidelity score:

$$s_{\text{fid}}^{(i)} = f_{\text{co}}(x^{(i)}).$$

Higher values indicate stronger agreement with the distribution of genuine co-morbid radiographs. We then define the binary fidelity-pass indicator

$$r_{\text{fid}}^{(i)} = \begin{cases} 1, & \text{if } s_{\text{fid}}^{(i)} \geq \tau_{\text{co}}, \\ 0, & \text{otherwise.} \end{cases}$$

This score is therefore intended as a proxy for radiological plausibility of the joint presentation, not merely for the separate presence of both labels.

5.3.4 Summary

Phase 3 transfers the representation hypothesis of Phase 2 to a clinical low-resource co-morbidity setting using VinBigData. It assumes that shared anatomy can be represented by z_c, disease-specific effects can be factorised from that shared context, and composition can therefore be performed over the generative variable u conditioned on z_c. To test this, Phase 3 combines a partitioned latent architecture, a factor-separating objective, and a single conditional latent diffusion model.

The central test is whether this representation supports held-out co-morbid composition. If composed samples recover both target pathologies, remain spatially plausible, and are judged consistent with real co-morbid radiographs, then Phase 3 supports H3. Otherwise, either the conditional assumptions are too weak in this clinical setting or the learned representation is still not sufficiently factorised to support stable composition.

Chapter 6

Research Plan

6.1 Time Plan

	2024					2025												2026												2027					
	08	09	10	11	12	01	02	03	04	05	06	07	08	09	10	11	12	01	02	03	04	05	06	07	08	09	10	11	12	01	02	03	04	05	06
Paper Reading																																			
Writing Proposal																																			
Phase 1																																			
Phase 2																																			
Phase 3																																			
Write Thesis																																			

6.2 Success Criteria

- Phase 1 should deliver a stable taxonomy benchmark and trajectory-based evidence that the composability gap grows predictably from structurally easy pairs to conflicting pairs.
- Phase 2 should show that better factorised latent representations improve held-out composition and reduce the gap between semantic and logical composition under a shared downstream evaluator.
- Phase 3 should produce VinBigData compositions that preserve both target pathologies, remain spatially plausible, and score as credible co-morbid radiographs under external evaluation.
- The full programme should end with reproducible manifests, evaluation scripts, and thesis-ready chapters that support publication and final submission.

6.3 Possible Challenges

- Diffusion training, shared-noise sweeps, and repeated ablations are computationally expensive and may slow iteration on limited hardware budgets.
- Composition quality can be sensitive to seeds, hyper-parameters, and representation choices, so multiple controlled runs are needed before drawing conclusions.
- Real co-morbid chest X-ray cases remain scarce, which limits the strength of supervision available for the clinical phase.
- Clinical validity depends partly on external classifiers and proxy realism checks, so failure analysis and specialist interpretation remain essential.

6.4 Limitations

The main limitation of this research lies in Phase 3, which depends on the selected pathology pair being spatially distinct enough for approximate conditional-independence assumptions and factor separation to hold. Diffusion models are also computationally expensive, limiting the scale of experiments and ablations. Finally, evaluation relies on external classifiers and proxy plausibility measures, while stronger validation would require radiologist assessment of the generated co-morbid radiographs.

6.5 Conclusion

This plan moves from characterising the composability gap, to testing whether better factorised representations reduce it, and finally to evaluating whether the same logic supports clinically plausible co-morbid chest X-ray synthesis. The phased structure keeps each stage tied to concrete outputs, making the overall programme feasible, cumulative, and coherent for thesis completion.

Chapter 7

Conclusion

This proposal argues that the main barrier to reliable compositional generation is the mismatch between logical composition and the semantically correct joint concept. To address this problem, the project proceeds in three stages: first, it characterises the composability gap through a structured benchmark and trajectory-based analysis; second, it tests whether more factorised latent representations can reduce this gap in a controlled setting; and third, it evaluates whether the same principles can support clinically plausible synthesis of rare co-morbid chest X-rays.

Taken together, the proposed work aims to contribute both a clearer theoretical account of why compositional diffusion models fail and a practical pathway for improving out-of-distribution generation in low-resource domains. If successful, the thesis will show that reliable composition depends not only on how conditions are combined at inference time, but also on how cleanly the underlying concepts are represented, with direct implications for generative modeling and medical image synthesis.

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